3 Cellular Energetics MCQ

1. A researcher claims that the incorporation of carbon dioxide into organic molecules during photosynthesis does not violate the second law of thermodynamics.

Which of the following statements best helps justify the researcher's claim?

(A) Organisms contain enzymes that lower the activation energies of specific chemical reactions.

(B) An ecosystem is formed by the interaction of a community of organisms with their surrounding environment.

(C) Photosynthetic organisms use the organic molecules produced during photosynthesis for growth and repair.

(D) The total system that includes photosynthetic organisms and the Sun becomes less ordered over time. ✓

Answer D

Correct. Because the total system that includes photosynthetic organisms and the Sun becomes less ordered over time, the incorporation of carbon dioxide molecules into organic molecules that occurs during photosynthesis does not violate the second law of thermodynamics.

2. Which two cellular organelles in eukaryotes have both electron transport systems and chemiosmotic mechanisms?

(A) Ribosomes and endoplasmic reticulum

(B) Chloroplasts and endoplasmic reticulum

(C) Chloroplasts and mitochondria ✓

(D) Mitochondria and nuclei

(E) Nuclei and Golgi apparatus

3. According to the chemiosmotic theory (chemiosmotic coupling), the energy required to move protons from the mitochondrial matrix to the intermembrane space against a concentration gradient comes most directly from

(A) photons of red or blue light

(B) the hydrolysis of ATP

(C) the breakdown of high-energy fatty acids in the mitochondrial matrix

(D) electrons flowing along the electron transport chain ✓

(E) substrate-level phosphorylation

3 Cellular Energetics MCQ

An experiment to measure the rate of respiration in crickets and mice at 10° C and 25° C was performed using a respirometer, an apparatus that measures changes in gas volume. Respiration was measured in mL of O_2 consumed per gram of organism over several five-minute trials, and the following data were obtained.

Organism	Temperature (°C)	Average respiration (mL O_2/g/min)
Mouse	10	0.0518
Mouse	25	0.0321
Cricket	10	0.0013
Cricket	25	0.0038

4. According to the data, the mice at 10° C demonstrated greater oxygen consumption per gram of tissue than did the mice at 25° C. This is most likely explained by which of the following statements?

(A) The mice at 10° C had a higher rate of ATP production than the mice at 25° C. ✓

(B) The mice at 10° C had a lower metabolic rate than the mice at 25° C.

(C) The mice at 25° C weighed less than the mice at 10° C.

(D) The mice at 25° C were more active than the mice at 10° C.

5. During aerobic cellular respiration, oxygen gas is consumed at the same rate as carbon dioxide gas is produced. In order to provide accurate volumetric measurements of oxygen gas consumption, the experimental setup should include which of the following?

(A) A substance that removes carbon dioxide gas ✓

(B) A plant to produce oxygen

(C) A glucose reserve

(D) A valve to release excess water

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(C) The mice at 25°C weighed less than the mice at 10°C.

(D) The mice at 25°C were more active than the mice at 10°C.

Answer A

This option is correct. Mice are endotherms and at cold temperatures will increase their rate of ATP production in order to shiver. This increase in metabolism will produce heat according to the second law of thermodynamics. This heat production will help the mouse maintain a constant internal environment.

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(D) A valve to release excess water

Answer A

This option is correct. There is no change in gas volume measured by the respirometer unless carbon

3 Cellular Energetics MCQ

dioxide is removed, because oxygen is consumed at the same rate that carbon dioxide is produced during cellular respiration.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

	Dish A		Dish B	
	Day 7 Covered	Day 14 Uncovered	Day 7 Uncovered	Day 14 Uncovered
Germinated seeds	12	20	20	20
Green-leaved seedlings	0	14	15	15
Yellow-leaved seedlings	12	6	5	5
Mean stem length below first set of leaves	8 mm	9 mm	3 mm	3 mm

8. According to the results of this experiment, germination of tobacco seeds during the first week is

(A) increased by exposure to light ✓

(B) unaffected by light intensity

(C) prevented by paper towels

(D) accelerated in green-leaved seedlings

Answer A

This option is correct. Dish B, which was uncovered, had 20 seeds germinate, while the covered dish A had only 12 seeds germinate in the first week.

9. Additional observations were made on day 21, and no yellow-leaved seedlings were found alive in either dish. This is most likely because

(A) yellow-leaved seedlings were unable to absorb water from the paper towels

(B) taller green-leaved seedlings blocked the light and prevented photosynthesis

(C) yellow-leaved seedlings were unable to convert light energy to chemical energy ✓

(D) a higher rate of respiration in yellow-leaved seedlings depleted their stored nutrients

3 Cellular Energetics MCQ

Answer C

This option is correct. The only difference between the two plants after day 14 was the leaf color. Without chlorophyll, the yellow seedlings could not absorb enough light energy to convert to chemical energy to sustain growth. Therefore, no yellow seedlings were found alive in either dish after day 21.

A student placed 20 tobacco seeds of the same species on moist paper towels in each of two petri dishes. Dish A was wrapped completely in an opaque cover to exclude all light. Dish B was not wrapped. The dishes were placed equidistant from a light source set to a cycle of 14 hours of light and 10 hours of dark. All other conditions were the same for both dishes. The dishes were examined after 7 days, and the opaque cover was permanently removed from dish A. Both dishes were returned to the light and examined again at 14 days. The following data were obtained.

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3 Cellular Energetics MCQ

12. Some cells release active signaling proteins when membrane-bound precursor proteins are cleaved by proteolytic enzymes. The signaling proteins can then bind to receptors on the surface of a target cell, thereby activating an intracellular signaling pathway and eliciting a response from the target cell.

This mechanism of activating receptor-binding signaling proteins has been observed in a variety of organisms from bacteria to humans. Many of the enzymes responsible for proteolysis of membrane-bound precursor proteins have been isolated and characterized.

Which of the following questions would be most appropriate to investigate whether the proteolytic enzymes are evolutionarily conserved among species?

(A) Are the genes encoding the proteolytic enzymes expressed in the same cell types in all species?

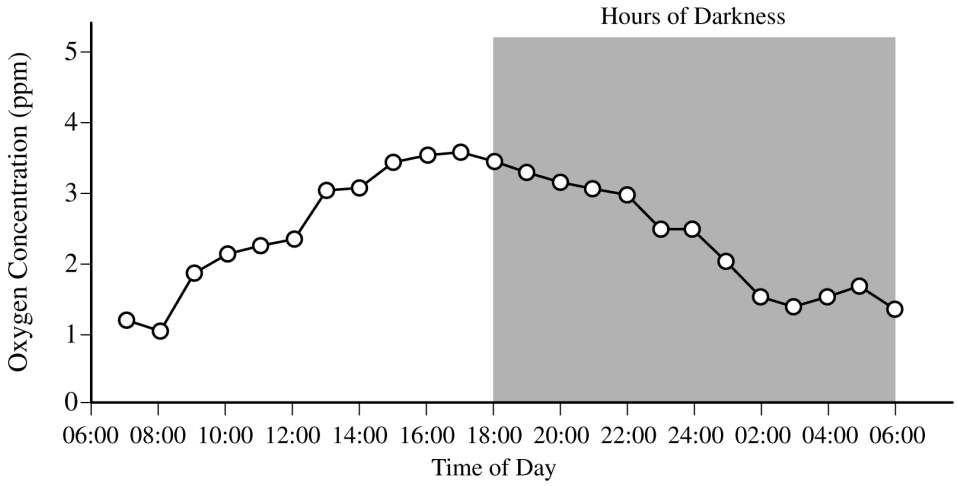
(B) Once the precursor proteins of different species are cleaved, do the active signaling proteins bind to the same receptors on different target cells?

(C) If a proteolytic enzyme from one species is incubated with a precursor protein from another species, does correct cleavage occur? ✓

(D) Are the proteolytic enzymes synthesized in the rough endoplasmic reticulum of all species?

13.

OXYGEN CONCENTRATION IN THE WATER OF A LAKE



What most likely causes the trends in oxygen concentration shown in the graph above?

(A) The water becomes colder at night and thus holds more oxygen.

(B) Respiration in most organisms increases at night.

(C) More organisms are respiring at night than during the day.

(D) Photosynthesis produces more oxygen than is consumed by respiration during the day. ✓

3 Cellular Energetics MCQ

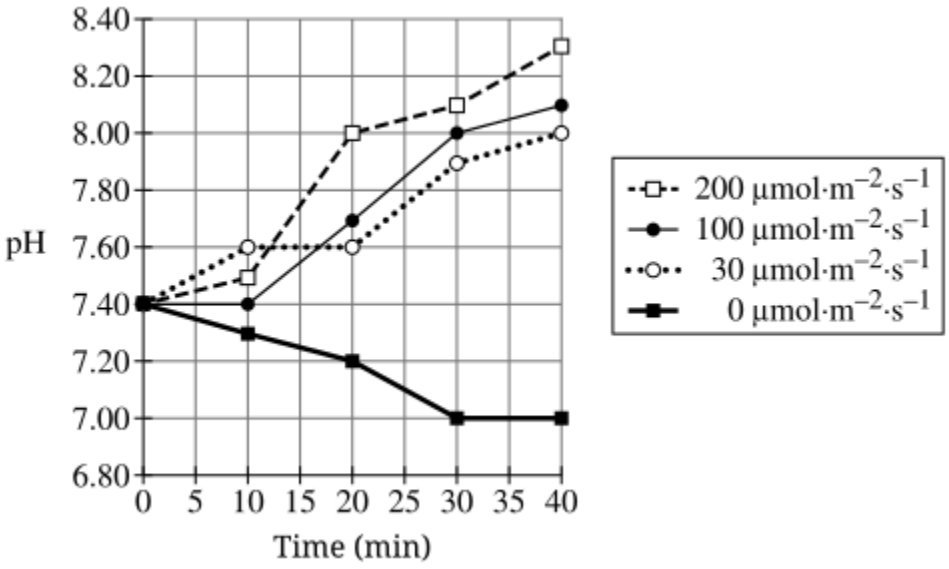


Figure 1. Effect of light intensity ($\mu mol \cdot m^{-2} \cdot s^{-1}$) on the pH of a solution containing algae

Students investigated the effect of light intensity on the rates of photosynthesis and cellular respiration in algae. They placed equal numbers of algal cells into test tubes containing a solution at $pH\ 7.40$. The test tubes were divided into four treatment groups, and each treatment group was exposed to a different light intensity. The pH in the tubes was measured every 10 minutes and used as an indicator of CO_2 concentration because CO_2 produces an acidic solution when dissolved in water (Figure 1).

14. Which of the following claims is best supported by the data?

(A) Algae only carry out photosynthesis when exposed to $200\ \mu mol \cdot m^{-2} \cdot s^{-1}$.

(B) Algae double the rate of photosynthesis when the light intensity doubles.

(C) Algae that are exposed to $30\ \mu mol \cdot m^{-2} \cdot s^{-1}$ of light perform photosynthesis instead of cellular respiration.

(D) Algae that are exposed to $100\ \mu mol \cdot m^{-2} \cdot s^{-1}$ of light have a greater rate of photosynthesis than of cellular respiration over time. ✓

Answer D

Correct. As indicated by the increase in pH, the rate of photosynthesis at $100\ \mu mol \cdot m^{-2} \cdot s^{-1}$ increased over time as the CO_2 produced by respiration was consumed in photosynthesis. This contrasts with the pH of the treatment group exposed to no light ($0\ \mu mol \cdot m^{-2} \cdot s^{-1}$) in which the pH decreased over time as CO_2 accumulated, because respiration but not photosynthesis was occurring.

3 Cellular Energetics MCQ

15. Scientists claim that changes in the rate of photosynthesis will be reflected by changes in pH.

Which of the following best supports their claim?

(A) Algae performing photosynthesis remove $\mathrm{CO_2}$ from the solution, increasing pH. ✓

(B) Algae performing photosynthesis release $\mathrm{CO_2}$ into the solution, decreasing pH.

(C) Algae performing photosynthesis remove $\mathrm{O_2}$ from the solution, decreasing pH.

(D) Algae performing photosynthesis release $\mathrm{O_2}$ into the solution, increasing pH.

Answer A

Correct. As compared with the "no light" control, the pH of each treatment group increased over time as the $\mathrm{CO_2}$ produced by respiration was consumed in photosynthesis. As the rate of photosynthesis increases, so does the pH.

16. Based on Figure 1, which of the following environmental changes will result in the greatest increase in the net photosynthetic productivity by the algae?

(A) A change in light intensity from $0\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ to $30\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ ✓

(B) A change in light intensity from $30\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ to $100\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$

(C) A change in light intensity from $100\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ to $200\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$

(D) A change in light intensity from $200\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ to $30\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$

Answer A

Correct. At the end of the experiment, the greatest increase in net photosynthetic productivity, as measured by a change in pH, was between $0\ \mu\mathrm{mol} \cdot \mathrm{m^{-2}} \cdot \mathrm{s^{-1}}$ and $30\ \mu\mathrm{mol} \cdot \mathrm{m^{-2}} \cdot \mathrm{s^{-1}}$. There was a 10-fold increase in pH between these two treatments, from 7.0 to 8.0.

17. Based on Figure 1, which of the following was used as a control in the experiment?

(A) A test tube placed directly in front of a light source

(B) The $0\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ light-intensity treatment ✓

(C) The $200\ \mu\mathrm{mol \cdot m^{-2} \cdot s^{-1}}$ light-intensity treatment

(D) A test tube containing no algal cells

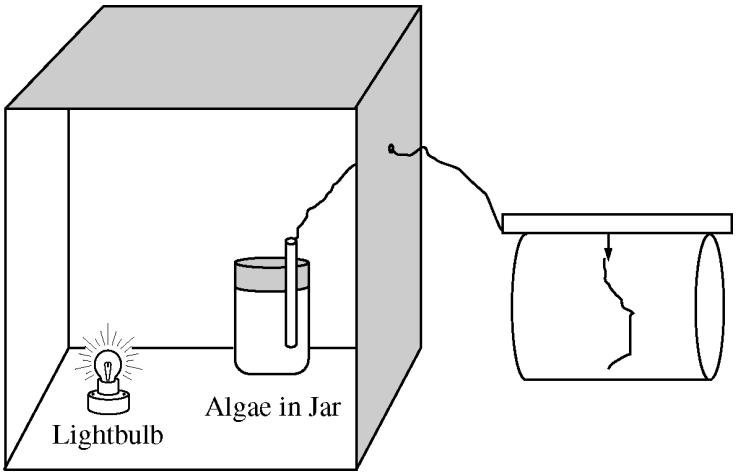
3 Cellular Energetics MCQ

Answer B

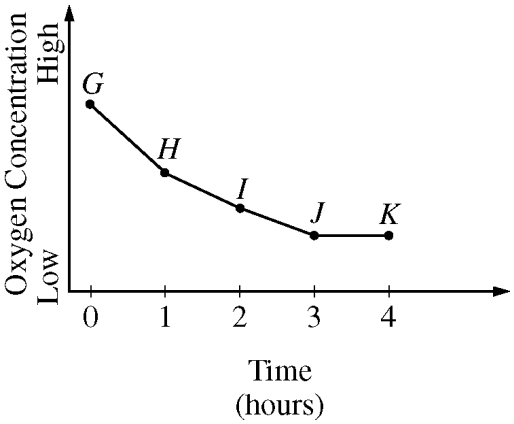
Correct. No light served as a control when measuring the effect of various light intensities.

For following group of questions first study the description of the situation and diagrams and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

A student studied the effects of light intensity on oxygen production in green algae. The algae were suspended in water inside a sealed glass jar, and the jar was placed into a constant-temperature, lightproof box containing a light source. A probe was inserted into the jar to record the concentration of oxygen. The probe was connected to a recording device. The setup is shown below.



The student decreased the intensity of the light hourly and recorded the corresponding changes in oxygen concentration. The graph below shows the results from the recording device.



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18. An increase in the rate of oxygen production by algae would be accompanied by a comparable increase in the rate of production of which of the following substances?

(A) $C_6H_{12}O_6$ ✓

(B) CO_2

(C) CH_4

(D) NH_3

(E) H_2O

19. Based on the data shown, changes in the light intensity resulted in changes in the rate of which of the following processes?

(A) Excretion

(B) Photosynthesis ✓

(C) Respiration

(D) Translation

(E) Transcription

20. The rate of oxygen production equaled the rate of oxygen consumption during which of the following time periods?

(A) *G* to *H*

(B) *H* to *I*

(C) *I* to *J*

(D) *J* to *K* ✓

(E) *G* to *K*

Directions: This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

This group of questions refers to molecules of the following substances.

(A) Cytochrome
(B) $FADH_2$
(C) NAD^+
(D) $NADP^+$
(E) Oxygen (O_2)

21. An intermediate electron acceptor for oxidations that occur in both glycolysis and in Krebs cycle reactions

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(A) Cytochrome
(B) $FADH_2$
(C) NAD^+ ✓
(D) $NADP^+$
(E) Oxygen (O_2)

22. Coenzyme that transfers electrons from the Krebs cycle to the mitochondrial electron-transport chain at a lower energy level than that of electrons entering at the beginning of the chain

(A) Cytochrome
(B) $FADH_2$ ✓
(C) NAD^+
(D) $NADP^+$
(E) Oxygen (O_2)

---

23. A researcher claims that increased atmospheric carbon dioxide levels cause increased growth rates in plants.

Which of the following statements best supports the researcher's claim?

(A) Atmospheric carbon dioxide is produced by the burning of fossil fuels, which are formed from the remains of living organisms such as plants.
(B) Atmospheric carbon dioxide is a byproduct of cellular respiration, which is a metabolic process that occurs in plants and other living organisms.
(C) Atmospheric carbon dioxide typically enters plant leaves through stomata, which plants rely on for regulating gas exchange with the atmosphere.
(D) Atmospheric carbon dioxide is the raw material for photosynthesis, which plants rely on for producing sugars and other organic compounds. ✓

Answer D

Correct. Atmospheric carbon dioxide is the raw material for photosynthesis, which is the metabolic process by which green plants use energy in sunlight to produce sugars and other organic compounds. The plants use the sugars and organic compounds to support growth, repair, and reproduction.

24. A researcher claims that the synthesis of ATP from ADP and inorganic phosphate $(\mathrm{P_i})$ is essential to cellular function.

Which of the following statements best helps justify the researcher's claim?

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(A) ADP is a small molecule that some cells release into their environment as a way of communicating with other cells.

(B) ATP hydrolysis is an energy-releasing reaction that is often coupled with reactions that require an input of energy. ✓

(C) Inorganic phosphate $(\mathrm{P_i})$ is a substance that cells typically acquire from their environment.

(D) ATP synthase is a mitochondrial enzyme that catalyzes the conversion of ADP and $\mathrm{P_i}$ to ATP.

Answer B

Correct. Many cellular processes necessary for maintaining the internal organization of the cell are coupled with the hydrolysis of ATP. Without a supply of ATP, these activities will stop and the cells will die.

25. During respiration, most ATP is formed as a direct result of the net movement of

(A) potassium against a concentration gradient

(B) protons down a concentration gradient ✓

(C) electrons against a concentration gradient

(D) electrons through a channel

(E) sodium ions into the cell

26. A scientist performed an experiment to test the ability of chloroplasts to produce ATP in the dark. The scientist maintained chloroplasts under dark conditions and placed them first in a pH 4 solution for 15 seconds and then in a pH 8 solution containing ADP and P_i (inorganic phosphate). Under these conditions, the pH of the stroma increased to 8, the pH of the thylakoid space remained at 4, and the chloroplasts produced ATP.

Which of the following best explains how the difference in pH between the stroma and the thylakoid space enabled the chloroplasts to produce ATP in the dark?

(A) The difference in pH created a proton gradient that enabled protons to pass through ATP synthase, activating it to produce ATP. ✓

(B) The difference in pH enabled complexes of protons and ADP in the thylakoid space to move through ATP synthase to the stroma, producing ATP.

(C) The difference in pH stimulated the electron flow needed to pump protons into the stroma and induce P_i and ADP to bond and form ATP.

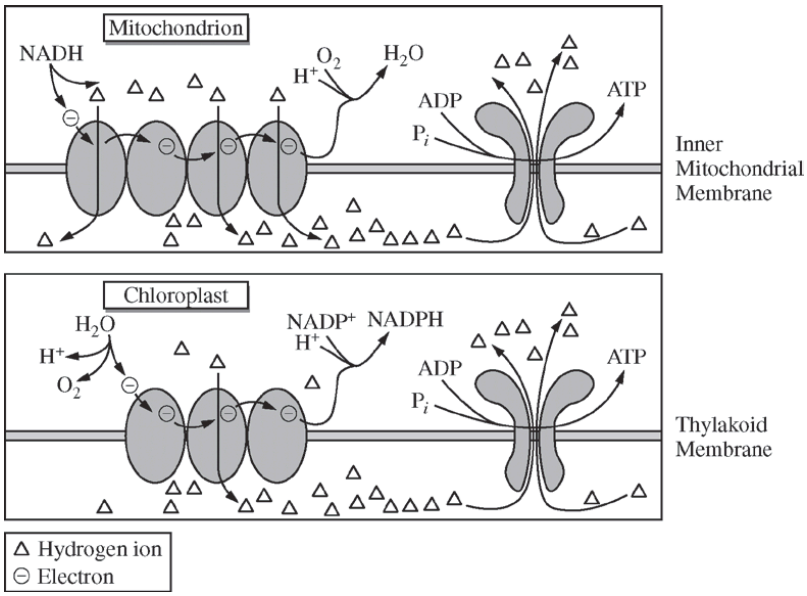
(D) The difference in pH caused the photosystems to release high energy electrons that provided enough energy for ADP and P_i to bond and form ATP.

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Answer A

Correct. Normal chloroplast function produces ATP by using photosystems to capture light, which generates high-energy electrons that are passed down an electron transport chain (ETC). Members of the ETC pump protons, creating a proton gradient between the thylakoid space and the stroma. Protons then move down their gradient through ATP synthase. The passage of protons provides energy to ATP synthase to form ATP from ADP and P_i. The scientist achieved this same effect by using low, then high, pH solutions in the absence of light.

27. The figures below illustrate the similarities between ATP synthesis in mitochondria and chloroplasts.



The figures can best assist in answering which of the following questions?

(A) Do electron transport chains create a gradient so that ATP synthase can generate ATP molecules? ✓

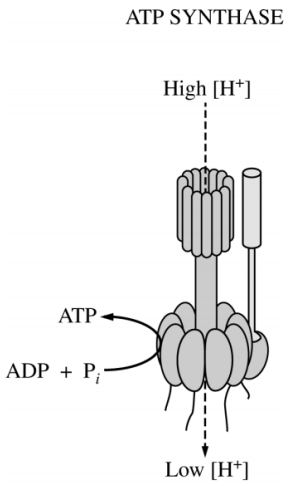
(B) What are the sources of energy that drive mitochondrial and chloroplast electron transport systems?

(C) What is the optimal temperature at which ATP synthase chemically converts ADP and a phosphate group into one molecule of ATP?

(D) What is the evolutionary relationship between the ATP synthase in mitochondria and the ATP synthase in chloroplasts?

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28.



Which of the following questions will best direct an investigation of the mechanism of ATP synthase?

(A) What is the source of the inorganic phosphate that is used to generate ATP from ADP?

(B) Is the phosphorylation of ADP by ATP synthase dependent on the formation of a proton gradient? ✓

(C) Can ATP synthase use the energy released by phosphorylation of ADP to pump protons against a concentration gradient?

(D) Can oxidative phosphorylation be uncoupled from the electron transport chain?

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29. Brown fat is a type of fat tissue found in hibernating mammals. Inside the mitochondria of these fat tissue cells, these mammals have an uncoupling protein embedded in the inner mitochondrial membrane. This uncoupling protein allows hydrogen ions to leak from the intermembrane space back into the mitochondrial matrix. Figure 1 shows details of the processes in the inner mitochondrial membrane.

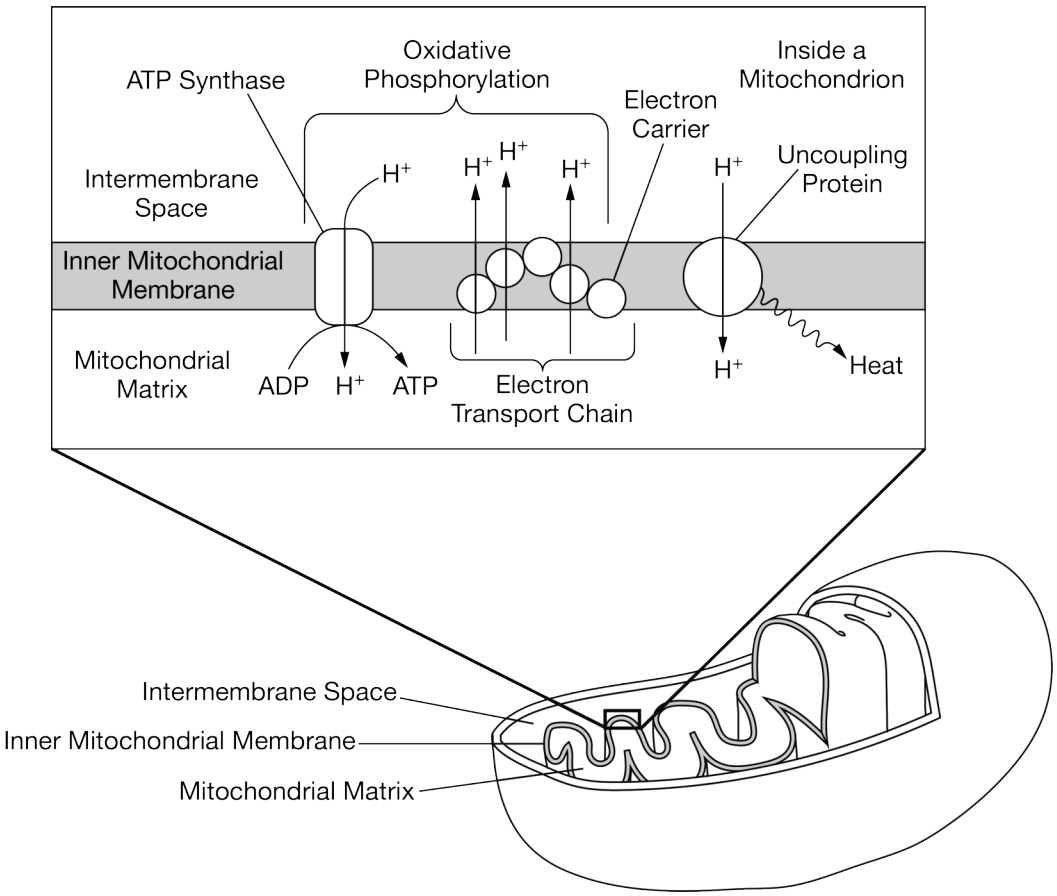


Figure 1. Processes in the inner mitochondrial membrane

Which of the following statements provides reasoning that supports the claim that brown fatty tissue keeps an animal warm?

(A) The uncoupling protein in this tissue increases the production of ATP and causes more body heat to be produced to warm the animal.

(B) The uncoupling protein in this tissue reduces the proton gradient across the membrane and thus produces heat to warm the animal without ATP production. ✓

(C) The uncoupling protein in this tissue causes an increase in the proton gradient, which causes more ATP to be produced that helps to warm the animal.

(D) The uncoupling protein in this tissue reduces the production of ATP and creates an increase in the proton gradient that allows more heat energy to be produced to warm the animal.

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Answer B

Correct. The uncoupling protein causes a reduction in the proton gradient across the inner mitochondrial membrane because protons constantly leak into the matrix. Heat energy is released as the protons move, but less ATP is made because of the decreased concentration of hydrogen ions available to move across ATP synthase.

30. The energy required to run the Calvin cycle reactions of photosynthesis comes from which two substances produced during the light-dependent reactions?

(A) ATP and NADPH ✓
(B) ADP and PO_4
(C) H^+ and PO_2
(D) O_2 and CO_2
(E) H_2O and CO_2

31. A researcher claims that only a portion of the light energy captured by green plants is available for growth and repair.

Which of the following observations best helps justify the researcher's claim?

(A) Light-capturing pigment molecules in green plants absorb red, blue, and violet light but reflect green light.
(B) The energy of a photon of light is proportional to its frequency and inversely proportional to its wavelength.
(C) As light energy is converted to chemical energy by metabolic processes, some of the energy is lost as heat. ✓
(D) Captured energy is stored in the molecular bonds of organic molecules, including simple sugars and starch.

Answer C

Correct. During photosynthesis, some of the captured light energy is lost to the environment in the form of heat energy. As a result, only some of the captured light energy is converted to chemical energy and stored for later use in growth and repair.

32. Carbohydrate-synthesizing reactions of photosynthesis directly require

3 Cellular Energetics MCQ

(A) light

(B) products of the light reactions ✓

(C) darkness

(D) O_2 and H_2O

(E) chlorophyll and CO_2

33. The carbon that makes up organic molecules in plants is derived directly from

(A) combustion of fuels

(B) carbon fixed in photosynthesis ✓

(C) carbon dioxide produced in respiration

(D) carbon in the lithosphere

(E) coal mines

34. Which of the following statements about mitochondrial chemiosmosis is NOT true?

(A) A proton gradient is established across the inner membrane of the mitochondrion.

(B) The potential energy released from the mitochondrial proton gradient is used to produce ATP.

(C) The mitochondrial proton gradient provides energy for muscle contraction.

(D) Proteins embedded in the inner mitochondrial membrane play an important role in ATP synthesis.

(E) Heat energy is required to establish the electron transport chain. ✓

35. Different photosynthetic organisms have different types of chlorophyll molecules. The distribution of chlorophylls in several different groups of organisms is shown in Table 1. A plus sign (+) in the table indicates the presence of a chlorophyll, while a minus sign (−) indicates its absence.

Table 1. The distribution of chlorophylls in several groups of organisms

	Chlorophyll *a*	**Chlorophyll *b***	**Chlorophyll *c***	**Chlorophyll *d***
Flowering plants	+	+	−	−
Green algae	+	+	−	−
Brown algae	+	−	+	−
Red algae	+	−	−	+
Cyanobacteria	+	−	−	−

Based on the data, which of the following most likely describes the evolutionary relationship among the organisms?

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(A) Because brown algae, red algae, and cyanobacteria lack chlorophyll *b*, they evolved before green algae and flowering plants did.

(B) Because green algae and flowering plants contain chloroplasts, they evolved more recently than brown algae, red algae, and cyanobacteria did.

(C) Because increasingly complex forms of chlorophyll are found in red algae, brown algae, green algae, and flowering plants, respectively, this reflects the order of their appearance.

(D) Because all of the organisms contain chlorophyll *a*, the organisms share a common ancestor. ✓

36. Which of the following best describes the function of the coenzymes NAD^+ and FAD in eukaryotic cellular respiration?

(A) They participate in hydrolysis reactions by accepting protons from water molecules.

(B) They participate directly in the phosphorylation of ADP to ATP.

(C) They serve as final electron acceptors in the electron transport chain.

(D) They aid vitamins such as niacin in the breakdown of glucose.

(E) They accept electrons during oxidation-reduction reactions. ✓

37. Within the cell, many chemical reactions that, by themselves, require energy input (have a positive free-energy change) can occur because the reactions

(A) may be coupled to the hydrolysis of ATP ✓

(B) take place very slowly

(C) take place when the cells are at unusually high temperatures

(D) are catalyzed by enzymes

(E) are aided by various metal ions that act as catalysts

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38. A researcher claims that the initial rise of oxygen in Earth's early atmosphere, which occurred approximately 2.3 billion years ago, resulted from the metabolic activity of prokaryotic organisms. The claim is based on an interpretation of the geochemical and fossil evidence represented in Figure 1.

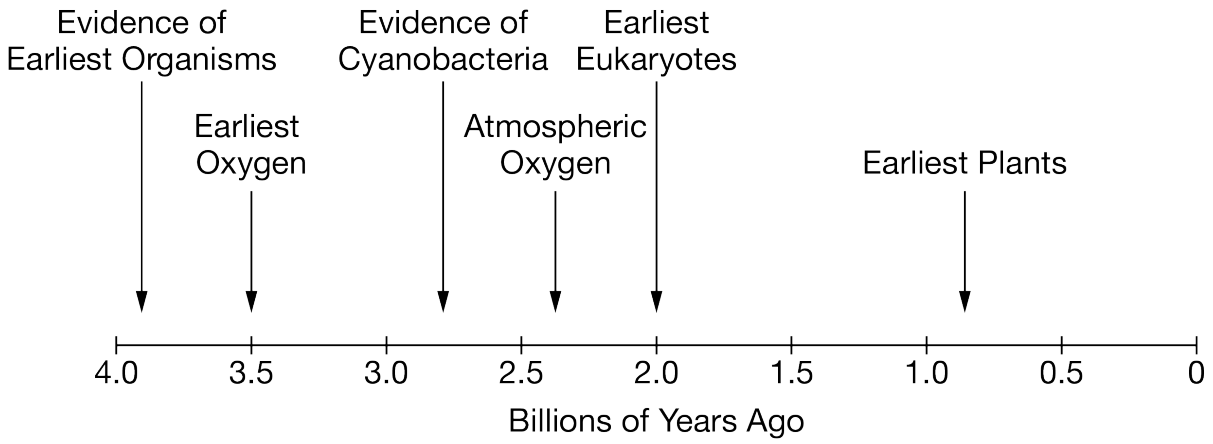


Figure 1. Selected events in geologic time based on geochemical and fossil evidence

Which of the following types of evidence will best support the researcher's claim?

(A) Evidence that some of the earliest eukaryotes used oxygen to produce ATP by cellular respiration

(B) Evidence that the earliest plants produced oxygen as a by-product of photosynthesis

(C) Evidence that some of the earliest organisms carried out photosynthesis without producing oxygen

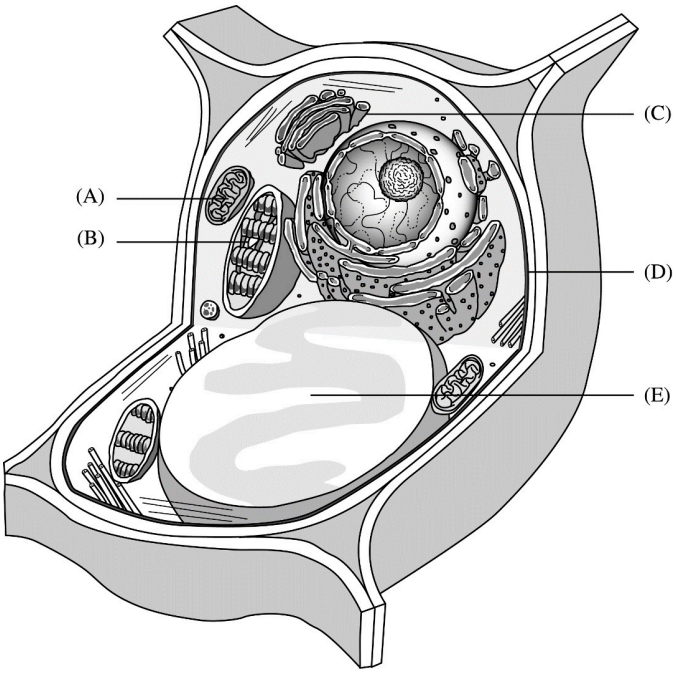
(D) Evidence that the cyanobacteria produced oxygen as a by-product of photosynthesis ✓

Answer D

Correct. Based on Figure 1, cyanobacteria appear in the fossil record before the rise of oxygen in Earth's early atmosphere. The researcher's claim will be supported by evidence that the cyanobacteria that were on Earth more 2.3 billion years ago produced oxygen as a by-product of photosynthesis. Also, cyanobacteria are prokaryotes, which is consistent with the researcher's claim.

3 Cellular Energetics MCQ

The following questions refer to the following diagram. For each phrase or sentence, select the labeled part to which it is most closely related. Each option may be used once, more than once, or not at all for each group.



39. Site of glucose synthesis

(A) A

(B) B ✓

(C) C

(D) D

(E) E

40. Site of conversion of chemical energy of glucose to ATP

(A) A ✓

(B) B

(C) C

(D) D

(E) E

3 Cellular Energetics MCQ

41.

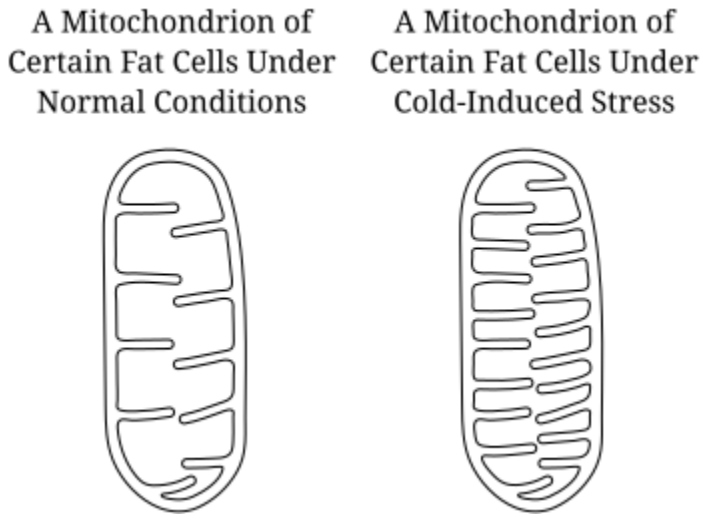


Figure 1. Model of a mitochondrion in cells under normal conditions and in cells experiencing cold-induced stress

In a certain type of fat cells, cold temperatures trigger a series of events that include changes to the structure of the mitochondria. A model of a mitochondrion from these fat cells under normal environmental conditions and when experiencing cold-induced stress is shown in Figure 1.

Which of the following best explains how the changes in mitochondrial structure observed when the cell experiences cold-induced stress affect cellular respiration?

(A) The volume of the mitochondrial intermembrane space decreases, preventing the reactions of the Krebs cycle from occurring.

(B) The surface area of the inner membrane decreases, preventing the transport of oxygen molecules into the mitochondria.

(C) The volume within the mitochondria increases, providing more space for the reactions of glycolysis to take place.

(D) The surface area of the inner membrane increases, providing more space for the components of the electron transport chain. ✓

Answer D

Correct. The increased folding of the inner membrane in cold-induced mitochondria results in an increase in surface area. This allows for more components of the electron transport chain (ETC) that is localized to the inner mitochondrial membrane. This in turn enables an increased rate of cellular respiration that can help maintain the body temperature of an endotherm exposed to cold temperatures.

3 Cellular Energetics MCQ

42.

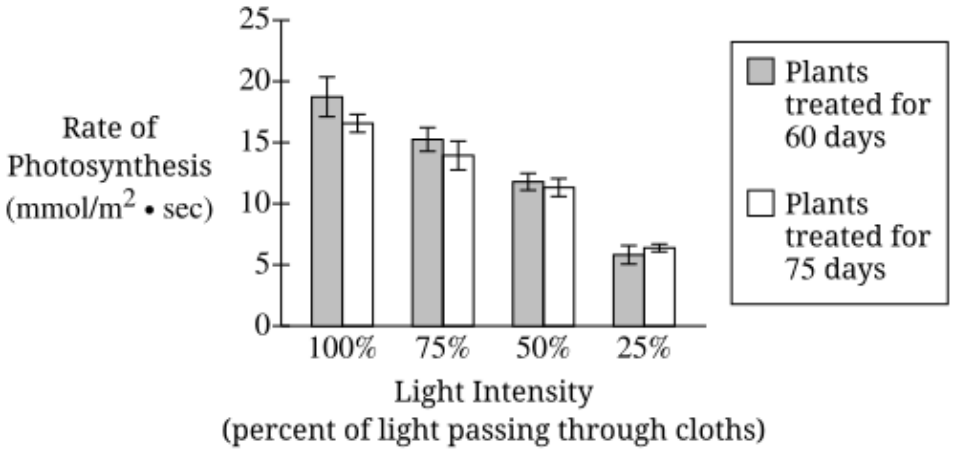


Figure 1. Effect of light intensity and growth period on the rate of photosynthesis in sage plants. Error bars represent $\pm 2\mathrm{SE}_{\bar{x}}$.

Using light-filtering cloths, scientists exposed sage plants to four different light intensities. Half of the sage plants were exposed to light for 60 days, while the other half were exposed to light for 75 days. The rate of photosynthesis in each plant was then determined (Figure 1).

Which of the following statements best describes the data?

(A) At 100% light intensity, the rate of photosynthesis was higher in plants that were treated for 60 days than in plants that were treated for 75 days.

(B) At 75% light intensity, the rate of photosynthesis was unaffected by an increase in treatment time. ✓

(C) At 50% light intensity, the rate of photosynthesis was the same as the rate of photosynthesis at 75% light intensity, regardless of the length of treatment time.

(D) At 25% light intensity, the rate of photosynthesis was higher in plants treated for 75 days than it was in plants treated for 60 days.

Answer B

Correct. Because the 60 days and 75 days error bars overlap, the rate of photosynthesis is the same for both treatments.

3 Cellular Energetics MCQ

43. Researchers conducted an experiment to investigate the effects of a valinomycin treatment on skeletal muscle cells. Valinomycin is a naturally occurring substance that can be used as a drug. The results of the experiment are presented in the table.

	Relative Rates of ATP Production	
Time after Treatment	Untreated Cells	Valinomycin-Treated Cells
5 minutes	1.0	0.3
10 minutes	7.7	2.7

Which of the following claims about the effects of the valinomycin treatment is best supported by the data presented in the table?

(A) The valinomycin treatment caused an increase in the activity of the rough endoplasmic reticulum.

(B) The valinomycin treatment caused an increase in the activity of the Golgi complex.

(C) The valinomycin treatment caused a decrease in the activity of the lysosome.

(D) The valinomycin treatment caused a decrease in the activity of the mitochondria. ✓

Answer D

Correct. The data indicate that the valinomycin treatment caused a decrease in the relative rate of ATP production, which likely resulted from impaired mitochondrial function.

44. Students in a class measured the mass of various living organisms. They then kept the organisms in the dark for 24 hours before remeasuring them. None of the organisms were provided with nutrients during the 24-hour period. The data are as follows.

Organism	Starting Mass (g)	Final Mass (g)
Elodea (submerged aquatic plant)	15.10	14.01
Goldfish	10.10	9.84
Sea anemone	25.60	24.98

Which of the following is the best explanation for the pattern of change in mass of the organisms over time?

3 Cellular Energetics MCQ

(A) Water loss due to evaporation

(B) Cellular respiration ✓

(C) The law of conservation of matter

(D) Growth and reproduction

45. The electron transport chain (ETC) is made up of several carrier molecules that transport electrons. These carrier molecules are found within membranes.

Which of the following statements best explains why these carrier molecules are typically found within membranes?

(A) The electrons that move through the ETC are directly used to drive the processes of endocytosis and exocytosis.

(B) The electrons that move through the ETC are coupled with the pumping of ATP out of the intermembrane space.

(C) The energy derived from the ETC creates a difference in the concentration of protons on either side of the membrane. ✓

(D) The energy derived from the ETC is used to fully oxidize an ATP molecule, releasing stored energy outside the membrane.

46.

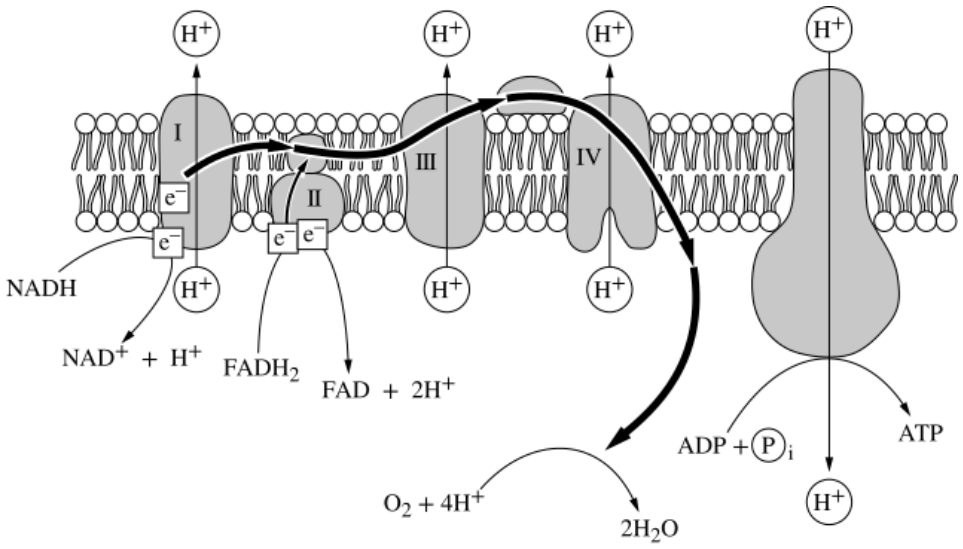


Figure 1. The electron transport chain of cellular respiration. The bold arrows passing through the complexes in the membrane represent the path of electron flow in mitochondria.

Compound X binds to complex IV of the mitochondrial electron transport chain and prevents complex IV from accepting electrons.

Based on Figure 1, which of the following best explains why the cells of an animal exposed to compound X have an increased ratio of $NADH$ to NAD^+?

3 Cellular Energetics MCQ

(A) $NADH$ increases because compound X directly prevents ATP synthase from producing the ATP necessary for the oxidation of $NADH$ to NAD^+.

(B) $NADH$ increases because the binding of compound X to complex IV also prevents the transport of NAD^+ out of the mitochondria.

(C) $NADH$ cannot be oxidized to NAD^+ because complexes I, II, and III cannot accept electrons if electrons cannot be passed to complex IV. ✓

(D) $NADH$ cannot be oxidized to NAD^+ because H_2O cannot bind to the active site of complex IV if this complex cannot accept electrons.

Answer C

Correct. If complex IV is blocked, complexes I, II, and III will remain reduced and unable to accept any more electrons. $NADH$ will be unable to transfer electrons to complex I and be oxidized to NAD^+, resulting in an increase in the ratio of $NADH$ to NAD^+ in the cells.

47. Experimental evidence shows that the process of glycolysis is present and virtually identical in organisms from all three domains, Archaea, Bacteria, and Eukarya. Which of the following hypotheses could be best supported by this evidence?

(A) All organisms carry out glycolysis in mitochondria.

(B) Glycolysis is a universal energy-releasing process and therefore suggests a common ancestor for all forms of life. ✓

(C) Across the three domains, all organisms depend solely on the process of anaerobic respiration for ATP production.

(D) The presence of glycolysis as an energy-releasing process in all organisms suggests that convergent evolution occurred.

48. Experimental evidence shows that the process of glycolysis is present and virtually identical in organisms from all three domains, Archaea, Bacteria, and Eukarya. Which of the following hypotheses could be best supported by this evidence?

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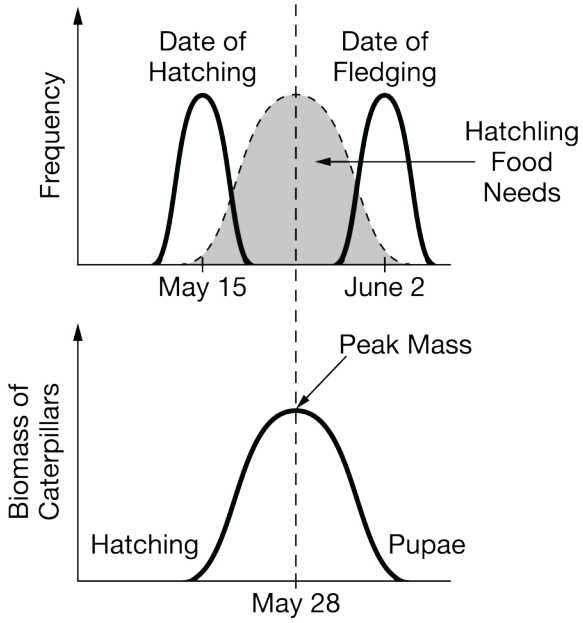
(B) Glycolysis is a universal energy-releasing process and therefore suggests a common ancestor for all forms of life. ✓

(C) Across the three domains, all organisms depend solely on the process of anaerobic respiration for ATP production.

(D) The presence of glycolysis as an energy releasing process in all organisms suggests that convergent evolution occurred.

3 Cellular Energetics MCQ

49. European flycatchers feed caterpillars to their hatchlings. Graph 1 shows the average dates of hatching and fledging (leaving the nest), and the biomass of the caterpillars between early May (when flycatcher young hatch) and June (when fledging of young occurs).



Graph 1. Comparison of European flycatcher hatching and fledging dates and caterpillar biomass

Based on the data, scientists claim that the reproductive behavior of European flycatchers is influenced by the availability of energy sources. Which of the following statements best justifies this claim?

(A) Young European flycatchers hatch from eggs when caterpillar biomass is available for the young birds to consume and convert into energy for growth. ✓

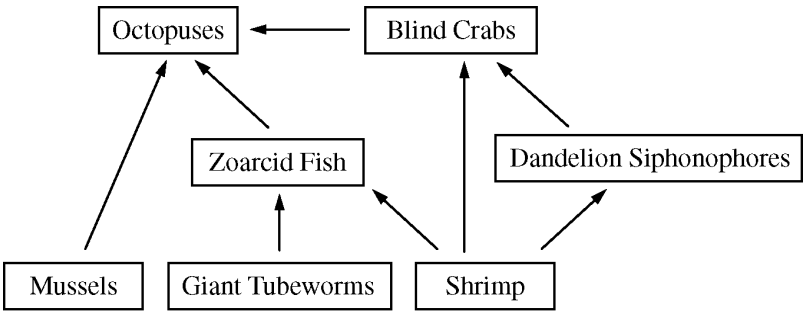
(B) European flycatcher hatchlings begin to need energy to leave the nest only after the caterpillars have turned into pupae.

(C) Female European flycatchers require energy to lay eggs, so they lay their eggs when the caterpillar biomass is maximal.

(D) The energy requirements for hatching European flycatchers and caterpillars are proportional to each other.

3 Cellular Energetics MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



The food web above represents feeding relationships in a biological community near a deep-sea hydrothermal vent. Hydrothermal vents are geysers on the seafloor that gush super-heated, mineral-rich water. The seawater surrounding hydrothermal vents typically contains carbon dioxide (CO_2), molecular hydrogen (H_2), hydrogen sulfide (H_2S), and methane (CH_4). Sunlight, however, fails to reach the seafloor where deep-sea hydrothermal vents are located.

As part of an investigation, researchers collected living specimens from an area near a deep-sea hydrothermal vent. Mussels in the collection were found to be dependent on molecular hydrogen in seawater. Also, the researchers discovered multiple species of bacteria living in the gills of the mussels. Mussels use gills for filter-feeding and gas exchange with the surrounding seawater. On the basis of their experimental results, the researchers hypothesized that some bacteria living in the gills of the mussels are capable of chemosynthesis.

50. Which of the following best explains how biological communities near deep-sea hydrothermal vents can exist in a habitat lacking sunlight?

(A) Environmental conditions on some distant planets resemble those experienced by organisms living near hydrothermal vents.

(B) Heterotrophs metabolize carbon-containing compounds produced by the photosynthetic organisms that live on the seafloor.

(C) Some organisms rely on energy captured from inorganic compounds to drive basic biological processes. ✓

(D) Some organisms that can tolerate high temperatures are single celled, whereas others are multicellular.

3 Cellular Energetics MCQ

51. A researcher claims that a certain herbicide suppresses plant growth by inhibiting chloroplast function. To test the claim, the researcher treats isolated chloroplasts with increasing concentrations of the herbicide. The data from the experiment are presented in Table 1.

Table 1. The effect of an herbicide on the function of isolated chloroplasts

Herbicide Concentration ($\mu g/L$)	pH Difference across Thylakoid Membrane	ATP Production
0	3.5	Very high
2	2.5	Moderate
8	0.5	Very low
32	0	Very low

Which of the following statements best clarifies how the data support the researcher's claim?

(A) The thylakoid membrane is more permeable to carbon dioxide than to polar molecules.

(B) ATP synthase activity depends on a proton gradient across the thylakoid membrane. ✓

(C) Some enzymes embedded in the thylakoid membrane catalyze the hydrolysis of ATP.

(D) Carbon fixation in the Calvin-Benson cycle takes place in the stroma of chloroplasts.

Answer B

Correct. The results of the experiment indicate that ATP production in the isolated chloroplasts decreases with increasing concentrations of the herbicide. The data also show that the decrease in ATP production is associated with a decrease in the pH difference across the thylakoid membrane. The pH difference is created by a proton gradient, which ATP synthase relies on to produce ATP.

52. Glycolysis is a metabolic pathway that converts glucose into pyruvate and is observed in each of the three domains. The hexokinase family of enzymes is required during glycolysis to phosphorylate six-carbon sugars. Researchers designed a general hexokinase inhibitor that is effective in the neurons of rats.

Which of the following best predicts the effect of adding this inhibitor to a culture of plant cells?

(A) Plant cells will be unaffected by the inhibitor as they do not perform glycolysis.

(B) Plant cells will be unable to perform glycolysis due to the inhibitor and will die. ✓

(C) Plant cells will be unable to perform photosynthesis due to the inhibitor and will die.

(D) Plant cells will still be able to perform glycolysis since plant hexokinase is not structurally similar to animal hexokinase.

3 Cellular Energetics MCQ

Answer B

Correct. The inhibitor is a general hexokinase inhibitor, which will prevent any form of hexokinase from functioning.

53. In a mesophyll cell of a leaf, the synthesis of ATP occurs in which of the following?

I. Ribosomes
II. Mitochondria
III. Chloroplasts

(A) I only
(B) II only
(C) III only
(D) II and III only ✓
(E) I, II, and III

54. In an experiment, a scientist isolates mitochondria from living cells and suspends them in two different buffered solutions. One solution is maintained at pH 4, while the other solution is maintained at pH 9. The scientist finds that mitochondria in the solution at pH 4 continue to produce ATP but those in the pH 9 solution do not.

The results of the experiment can be used as evidence in support of which of the following scientific claims about mitochondrial activity?

(A) Mitochondria in a cell-free environment are unable to convert thermal energy into ATP.
(B) The electron transport chain pumps electrons from the cytosol to the mitochondrial matrix.
(C) ATP production in mitochondria requires a hydrogen ion gradient that favors movement of protons into the mitochondrial matrix. ✓
(D) ATP synthase molecules change their orientation in relation to the proton gradient across the mitochondrial membrane.

55. In most freshwater fish, nitrogenous waste is primarily excreted as ammonia, which is highly soluble in water and is toxic at low concentrations. In terrestrial mammals, ammonia is converted to urea before it is excreted. Urea is also highly soluble in water but is less toxic than ammonia at low concentrations. Which of the following best explains why freshwater fish do <u>not</u> convert ammonia to urea for excretion?

(A) The metabolic pathways of fish do not normally involve nitrogen consumption.
(B) The dilution of ammonia by direct excretion into freshwater conserves energy. ✓
(C) Ammonia is concentrated in tissues, where it is stored prior to excretion.
(D) The nitrogen in ammonia is recycled for use in protein and nucleotide synthesis.

3 Cellular Energetics MCQ

Answer B

This option is correct. There is no need to convert ammonia to urea because it can be excreted often in dilute, nontoxic concentrations. This process requires less energy than the additional step of urea conversion. Dehydration due to water loss is not a problem for freshwater fish because water is plentiful.

56. Which of the following statements is true about the Krebs (citric acid) cycle and the Calvin (light-independent) cycle?

(A) They both result in a net production of ATP and NADH.
(B) They both require a net input of ATP.
(C) They both result in a release of oxygen.
(D) They both take place within the cytoplasmic matrix.
(E) They both are carried out by enzymes located within an organelle matrix. ✓

57. Which of the following best explains why the inner mitochondrial membrane has more embedded proteins of the electron transport chain than the outer mitochondrial membrane?

(A) The inner membrane is more efficient at capturing light than the outer membrane.
(B) The inner membrane has a smaller surface area than the outer membrane.
(C) The Krebs cycle occurs in the mitochondrial matrix and not the intermembrane space. ✓
(D) The outer membrane has a greater surface area for electron transfer.

Answer C

Correct. Since the Krebs cycle occurs in the mitochondrial matrix, having the electron transport chain in the inner membrane is more efficient as it allows direct access to the NADH and \(FADH_2\) produced by the Krebs cycle.

58. In chloroplasts, ATP is synthesized from ADP plus inorganic phosphate $(\mathrm{P_i})$ in a reaction catalyzed by ATP synthase molecules that are embedded in the thylakoid membrane.

Which of the following statements provides evidence to support the claim that no ATP will be synthesized in the absence of a proton gradient across the thylakoid membrane?

3 Cellular Energetics MCQ

(A) Blocking electron flow from one carrier to the next in the electron transport chains blocks formation of a proton gradient in the thylakoid.

(B) Increasing the proton concentration difference across the thylakoid membrane is not associated with a parallel increase in the rate of ATP synthesis.

(C) No ATP is synthesized when channel proteins that allow the free passage of protons are inserted into the thylakoid membrane. ✓

(D) No ATP is synthesized while the Calvin cycle is synthesizing carbohydrates and using ATP and NADPH at a high rate.

Answer C

Correct. Channel proteins that allow the passage of protons will dissipate the proton concentration difference, and ATP will not form. Under normal circumstances, the only way for protons to cross the thylakoid membrane is by passing through the ATP synthase molecules.

59. Which of the following is an important difference between light-dependent and light-independent reactions of photosynthesis?

(A) The light-dependent reactions occur only during the day; the light-independent reactions occur only during the night.

(B) The light-dependent reactions occur in the cytoplasm; the light-independent reactions occur in chloroplasts.

(C) The light-dependent reactions utilize CO_2 and H_2O; the light-independent reactions produce CO_2 and H_2O.

(D) The light-dependent reactions depend on the presence of both photosystems I and II; the light-independent reactions require only photosystem I.

(E) The light-dependent reactions produce ATP and NADPH; the light-independent reactions use energy stored in ATP and NADPH. ✓

60. Which of the following describes a metabolic consequence of a shortage of oxygen in muscle cells?

(A) An increase in blood pH due to the accumulation of lactic acid

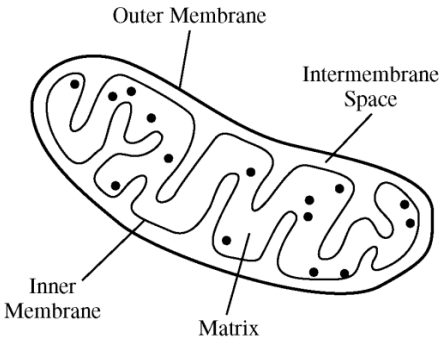
(B) No ATP production due to the absence of substrate-level phosphorylation

(C) A buildup of lactic acid in the muscle tissue due to fermentation ✓

(D) A decrease in the oxidation of fatty acids due to a shortage of ATP

3 Cellular Energetics MCQ

61.



The figure above shows an organelle typically found in eukaryotic cells. Which of the following best describes the function of the double membrane system of this organelle?

(A) The outer membrane allows the transport of all molecules into the intermembrane space, while the inner membrane serves as the regulatory boundary.

(B) The inner membrane has specialized proteins that create a hydrogen ion concentration gradient between the intermembrane space and the matrix. ✓

(C) The outer membrane contains transport proteins that establish a sodium ion concentration gradient used for ATP production, while the inner membrane contains transport proteins that establish a hydrogen ion concentration gradient used for glucose production.

(D) The toxins and wastes entering a cell cross the outer membrane and are detoxified by digestive enzymes stored within the intermembrane space.

62. In an experiment, researchers compared the growth of two different plants, plant X and plant Y. The researchers maintained the plants under nearly identical conditions and observed that plant X grew faster than plant Y. The researchers also observed that the inner mitochondrial membranes of plant X had more folds than did those of plant Y.

Which of the following conclusions about increasing the number of folds in the inner mitochondrial membrane is best supported by the results of the experiment?

(A) It increases the efficiency of photosynthesis, which results in faster cell growth.

(B) It increases the surface area available for ATP production, which results in faster cell growth. ✓

(C) It increases the amount of space available for storing cellular wastes, which results in faster cell growth.

(D) It increases the rate of protein transport to the plasma membrane, which results in faster cell growth.

Answer B

Correct. The increased surface area of the folds will contain more ATP synthase, allowing for more efficient use of the chemiosmotic gradient and more efficient production of ATP. The observation that plant X grew faster than plant Y supports this conclusion.

3 Cellular Energetics MCQ

63. More than 90 percent of the nitrogenous waste that is processed and excreted by humans is derived from the breakdown of proteins. Most of the remaining nitrogenous waste material is derived from the breakdown of

(A) fats

(B) plant cell walls

(C) steroids

(D) nucleic acids ✓

(E) glycogen

64. Muscle contraction depends on ATP hydrolysis. During periods of intense exercise, muscle cells rely on the ATP supplied by three metabolic pathways: glycolysis, mitochondrial respiration, and the phosphagen system. Figure 1 shows the rates at which the three metabolic pathways produce ATP following the start of an intense period of exercise.

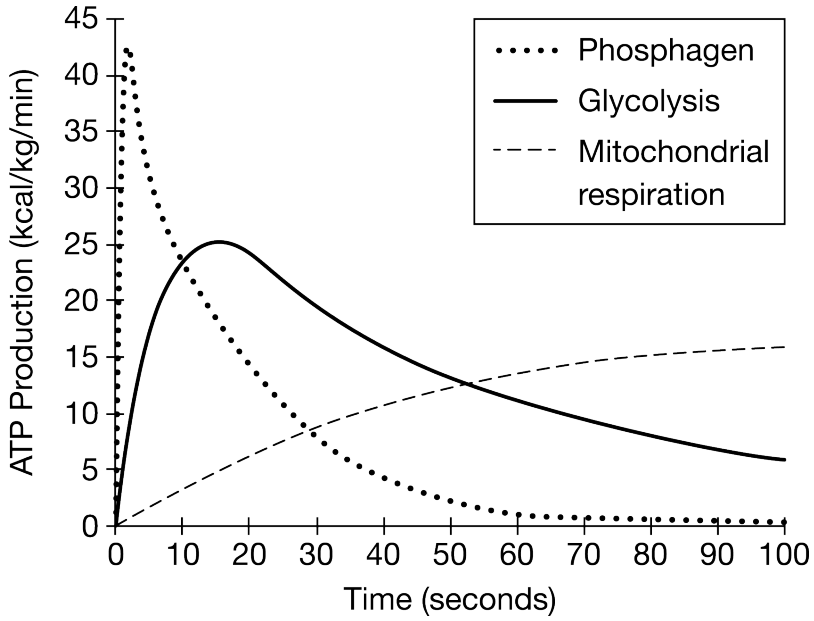


Figure 1. ATP production by three metabolic pathways following the start of an intense period of exercise

Which of the following correctly uses the data to justify the claim that the phosphagen system is an immediate, short-term source of ATP for muscle cells?

(A) ATP production by the phosphagen system increases and decreases rapidly following the start of the exercise period. ✓

(B) ATP production by the phosphagen system increases gradually and continuously throughout the entire exercise period.

(C) The ATP produced by the phosphagen system contains more energy per molecule than does the ATP produced by the other pathways.

(D) ATP hydrolysis in muscle cells occurs immediately after the start of the exercise period but stops before the end of the exercise period.

3 Cellular Energetics MCQ

Answer A

Correct. Based on Figure 1, the phosphagen system reaches its maximum rate of ATP production approximately 5 seconds after the start of the exercise period. Because ATP production by the phosphagen system increases and decreases rapidly following the start of the exercise period, the phosphagen system is an immediate, short-term source of ATP.

65. Two nutrient solutions are maintained at the same pH. Actively respiring mitochondria are isolated and placed into each of the two solutions. Oxygen gas is bubbled into one solution. The other solution is depleted of available oxygen. Which of the following best explains why ATP production is greater in the tube with oxygen than in the tube without oxygen?

(A) The rate of proton pumping across the inner mitochondrial membrane is lower in the sample without oxygen. ✓

(B) Electron transport is reduced in the absence of a plasma membrane.

(C) In the absence of oxygen, oxidative phosphorylation produces more ATP than does fermentation.

(D) In the presence of oxygen, glycolysis produces more ATP than in the absence of oxygen.

66. The function of which of the following organelles directly requires oxygen?

(A) Ribosome

(B) Mitochondrion ✓

(C) Nucleus

(D) Centriole

(E) Golgi apparatus

67. Which of the following best supports the claim that organisms of different domains share a common ancestor?

(A) Photosynthesis occurs in chloroplasts, and the citric acid cycle occurs in mitochondria.

(B) Glycolysis occurs in both prokaryotic cells and eukaryotic cells. ✓

(C) Introns are present in eukaryotic DNA but not in prokaryotic DNA.

(D) Errors in DNA synthesis provide some of the genetic variation in a population.

68. Which of the following statements best supports the claim that organisms share fundamental processes as a result of evolution?

3 Cellular Energetics MCQ

(A) All organisms that are introduced into new environments have the capacity to fill vacant ecological roles.

(B) All organisms have the ability to utilize oxygen to harness energy from the chemical breakdown of organic compounds.

(C) All organisms share a genetic code organized into triplet codons, making it possible for one organism to express a gene from another organism. ✓

(D) All organisms possess structures such as chloroplasts and mitochondria within their cells that reflect past symbiotic relationships between prokaryotic precursors.

Answer C

This option is correct. It demonstrates an understanding that all organisms, both extant and extinct, share a universal genetic code with the justification that the hereditary information encoded in DNA directs the development of biochemical, anatomical, and physiological traits by the fundamental processes of replication, transcription, and translation. Through the process of transformation, genetic information from one organism can change the genotype and phenotype of another organism, thus providing additional evidence for the universality of the genetic code.

69. Oxygen consumption can be used as a measure of metabolic rate because oxygen is

(A) necessary for ATP synthesis by oxidative phosphorylation ✓

(B) necessary to replenish glycogen levels

(C) necessary for fermentation to take place

(D) required by all living organisms

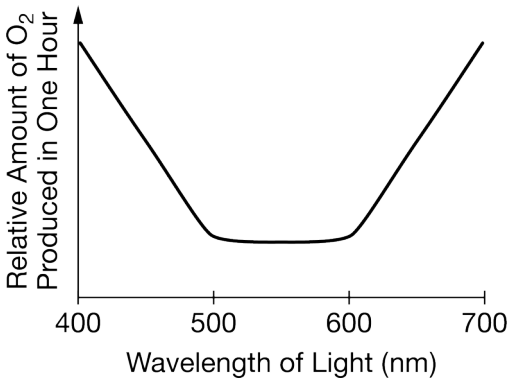
(E) required to break down the ethanol that is produced in muscles

70. A researcher claims that spinach leaves capture the most energy from light waves in the range of 500 nm to 600 nm. To test the claim, the researcher will place spinach leaves in separate chambers and expose the leaves to different wavelengths of light. For each chamber, the researcher will measure the amount of oxygen gas (O_2) that is produced in one hour.

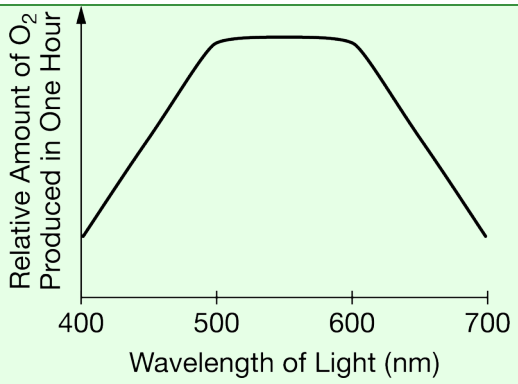
Which of the following graphs best represents data from the experiment that will support the researcher's claim?

3 Cellular Energetics MCQ

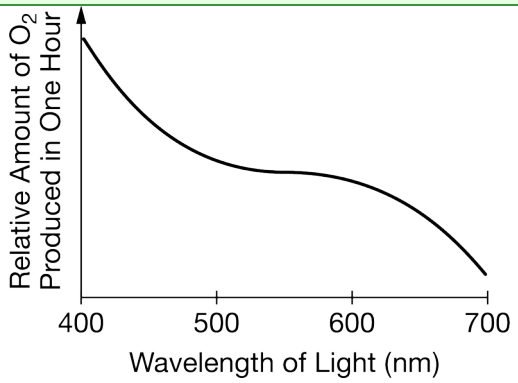
(A)



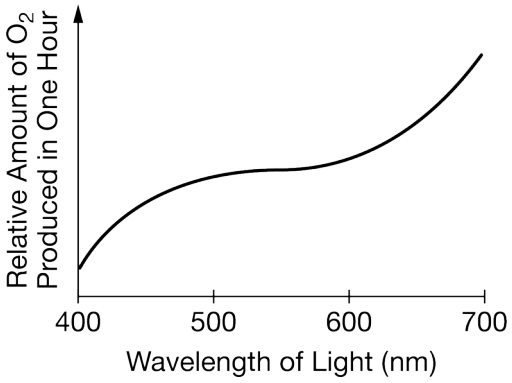
(B) ✓



(C)



(D)



3 Cellular Energetics MCQ

Answer B

Correct. The high levels of O_2 produced at wavelengths between 500 nm and 600 nm directly support the researcher's claim. Because oxygen is produced as a by-product of photosynthesis, a greater level of oxygen production indicates a greater level of energy-capture from light.

71. It is estimated that oxygen production first evolved in photosynthetic prokaryotes approximately 2.7 billion years ago. The first photosynthetic prokaryotes are presumed to be similar to today's cyanobacteria.

Which of the following best supports the claim that photosynthetic prokaryotes were responsible for the oxygen in Earth's atmosphere?

(A) The light reactions of photosynthesis split carbon dioxide into carbon and oxygen.

(B) The light reactions of photosynthesis split water into hydrogen ions and oxygen. ✓

(C) The Calvin cycle splits glucose into carbon, hydrogen, and oxygen.

(D) The Calvin cycle splits water into hydrogen ions and oxygen.

72. A scientist claims that *Elysia chlorotica,* a species of sea slug, is capable of photosynthesis.

Which of the following observations provides the best evidence to support the claim?

(A) *Elysia chlorotica* will die if not exposed to light.

(B) *Elysia chlorotica* grows when exposed to light in the absence of other food sources. ✓

(C) *Elysia chlorotica* grows faster when exposed to light than when placed in the dark.

(D) *Elysia chlorotica* grows in the dark when food sources are available.

Answer B

Correct. Growth requires an energy source. If the only energy source available is light, then the growth must be the result of photosynthesis capturing light energy for use by the sea slug.

73. To test the hypothesis that a particular plant synthesizes storage lipids by using glyceraldehyde 3-phosphate (G3P) from photosynthesis, a researcher plans to use radiolabeled precursors to track the molecules through the biosynthetic pathway. Which of the following radiolabeled precursors is most appropriate for the researcher to use?

3 Cellular Energetics MCQ

(A) ^{15}N-labeled N_2, because atmospheric nitrogen is fixed to amino acids by photosynthesis

(B) ^{14}C-labeled CO_2, because atmospheric carbon is fixed to carbohydrates by photosynthesis ✓

(C) ^{35}S-labeled methionine, because amino acids are incorporated into lipids during photosynthesis

(D) ^{32}P-labeled phosphate, because lipids are stored in plants as phospholipids

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- **Directions:** This group of questions consists of five lettered headings followed by a list of phrases or sentences. For each phrase or sentence, select the one heading to which it is most closely related. Each heading may be used once, more than once, or not at all.

(A) Glysolysis
(B) Krebs cycle (citric acid cycle)
(C) Calvin cycle (light-independent reactions of photosynthesis)
(D) Light-dependent reactions of photosynthesis
(E) Chemiosmosis

74. Process found in both photosynthesis and cellular respiration

(A) A
(B) B
(C) C
(D) D
(E) E ✓

75. Process in which carbon from CO_2 is incorporated into organic molecules

(A) A
(B) B
(C) C ✓
(D) D
(E) E

76. Process in which CO_2, is released as a by-product of oxidation-reduction reactions

(A) A
(B) B ✓
(C) C
(D) D
(E) E

77. Process in which O_2 is released as a by-product of oxidation-reduction reactions

3 Cellular Energetics MCQ

(A) A
(B) B
(C) C
(D) D ✓
(E) E

78. Process in which sugar is oxidized to pyruvic acid

(A) A ✓
(B) B
(C) C
(D) D
(E) E

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79. Some organisms use a weaker terminal electron acceptor than oxygen during cellular respiration. Which of the following best explains why the cells of these organisms might generate less ATP per molecule of glucose than cells that use oxygen as the terminal electron acceptor?

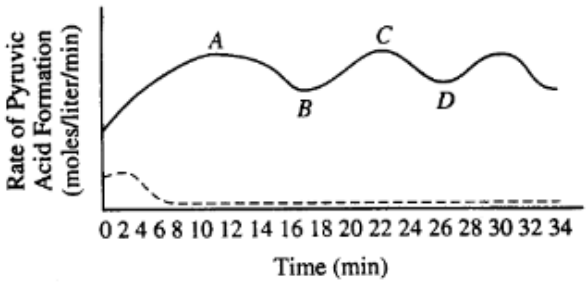
(A) The cells of these organisms remove fewer electrons from glucose than cells that use oxygen.
(B) The cells of these organisms create a smaller proton gradient than cells that use oxygen. ✓
(C) The cells of these organisms convert pyruvate to lactate more often than cells that use oxygen.
(D) The cells of these organisms generate more metabolic heat than cells that use oxygen.

80. When hydrogen ions are pumped out of the mitochondrial matrix, across the inner mitochondrial membrane, and into the space between the inner and outer membranes, the result is

(A) damage to the mitochondrion
(B) the reduction of NAD
(C) the restoration of the Na-K balance across the membrane
(D) the creation of a proton gradient ✓
(E) the lowering of pH in the mitochondrial matrix

3 Cellular Energetics MCQ

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.



A tissue culture of vertebrate muscle was provided with a constant excess supply of glucose under anaerobic conditions starting at time zero and the amounts of pyruvic acid and ATP produced were measured. The solid line in the graph above represents the pyruvic acid produced in moles per liter per minute. ATP levels were also found to be highest at points *A* and *C*, lowest at *B* and *D*. A second culture was set up under the same conditions, except that substance **X** was added, and the results are indicated by the dotted line.

81. Which of the following is most likely to result if oxygen is added to the tissue culture?

(A) Lactic acid formation will increase.

(B) For each glucose molecule consumed, more ATP will be formed. ✓

(C) The levels of ATP produced will decrease.

(D) Ethyl alcohol will be produced.

(E) No change in the production of pyruvic acid will be observed.

82. It is most reasonable to hypothesize that, in the breakdown of glucose, substance **X** is

(A) an activator

(B) an inhibitor ✓

(C) a substrate

(D) a coenzyme

(E) a cofactor

83. Which of the following best accounts for the shape of the solid line between points *A* and *D*?

(A) After ten minutes the cellular enzymes became ineffective.

(B) Respiration became uncontrolled.

(C) ATP acted as an allosteric inhibitor on one or more of the enzymes. ✓

(D) The measurements of pyruvic acid were unreliable.

(E) The cells required more glucose than was being provided.

3 Cellular Energetics MCQ

84. Which of the following questions is most relevant to understanding the Calvin cycle?

(A) How does chlorophyll capture light?

(B) How is ATP used in the formation of 3-carbon carbohydrates? ✓

(C) How is $NADP^+$ reduced to NADPH?

(D) How is ATP produced in chemiosmosis?

85. The chemical reaction for photosynthesis is

$6\ CO_2 + 12\ H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6\ O_2 + 6\ H_2O$

If the input water is labeled with a radioactive isotope of oxygen, ^{13}O, then the oxygen gas released as the reaction proceeds is also labeled with ^{13}O. Which of the following is the most likely explanation?

(A) During the light reactions of photosynthesis, water is split, the hydrogen atoms combine with the CO_2, and oxygen gas is released.

(B) During the light reactions of photosynthesis, water is split, removing electrons and protons, and oxygen gas is released. ✓

(C) During the Calvin cycle, water is split, regenerating NADPH from $NADP^+$, and oxygen gas is released.

(D) During the Calvin cycle, water is split, the hydrogen atoms are added to intermediates of sugar synthesis, and oxygen gas is released.

Answer B

This option is correct. This choice correctly describes the events of the light- dependent reactions and indicates that the student can justify the selection of data regarding the types of molecules that an organism will take up as necessary building blocks.

86. The chemical reaction for photosynthesis is

$6CO_2 + 12H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$

If the input water is labeled with a radioactive isotope of oxygen, ^{18}O, then the oxygen gas released as the reaction proceeds is also labeled with ^{18}O. Which of the following is the most likely explanation?

(A) During the light reactions of photosynthesis, water is split, the hydrogen atoms combine with the CO_2, and oxygen gas is released.

(B) During the light reactions of photosynthesis, water is split, removing electrons and protons, and oxygen gas is released. ✓

(C) During the Calvin cycle, water is split, regenerating NADPH from $NADP^+$, and oxygen gas is released.

(D) During the Calvin cycle, water is split, the hydrogen atoms are added to intermediates of sugar synthesis, and oxygen gas is released.

3 Cellular Energetics MCQ

87. The reactions of glycolysis occur in the

(A) cytosol ✓

(B) nucleus

(C) matrix of the mitochondrion

(D) membranes of the mitochondrion

(E) stroma of the chloroplast

88. Which metabolic process is common to both aerobic cellular respiration and alcoholic fermentation?

(A) Krebs cycle

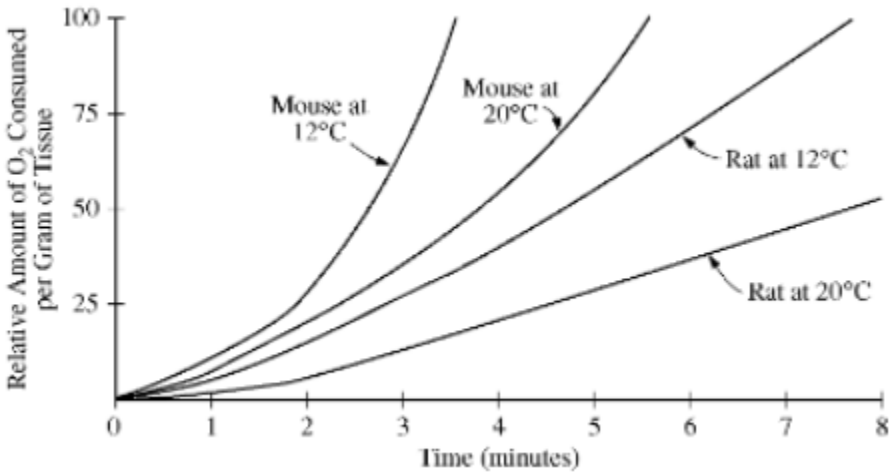
(B) Glycolysis ✓

(C) Electron transport chain

(D) Conversion of pyruvic acid to acetyl CoA

(E) Production of a proton gradient

A respirometer is a container used to measure the amount of oxygen consumed by an organism. A respirometer was used to determine how environmental temperature affects the uptake of oxygen in one 300-gram rat and one 50-gram mouse. The results of this experiment are shown on the graph below.



89. Which of the following hypotheses is best supported by the results of this experiment?

(A) Metabolic rate per gram of tissue is higher in smaller mammals. ✓

(B) Metabolic rate per gram of tissue is not related to body mass.

(C) Mice produce less CO_2 per gram of tissue than do rats.

(D) Rats consume more food per gram of tissue than do mice.

(E) Rats lose more heat per gram of tissue than do mice.

3 Cellular Energetics MCQ

90. Which of the following would most adversely affect the validity of this experiment?

(A) Large variation in metabolic rates between individuals of a species ✓

(B) Repeating the experiment several times

(C) Using a brown rat and a white mouse

(D) Using a thermometer that consistently read 2°C too low

(E) Using a scale that consistently indicated 2 grams too much

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91. Which of the following best describes the role of ATP and NADPH in the process of photosynthesis?

(A) ATP is required for the chemiosmotic production of NADPH during photosynthesis.

(B) ATP and NADPH are necessary to initiate the Krebs cycle in photosynthesis.

(C) ATP and NADPH are carriers of chemical energy that is used in the production of carbohydrates. ✓

(D) ATP and NADPH are required to boost electrons to a higher energy state in photosystems I and II.

Answer C

Correct. The energy captured from light during the photosynthetic light-dependent reactions is stored in ATP and NADPH for use in the Calvin cycle when carbon dioxide is fixed to enable the production of carbohydrates.

92. The FtsZ protein is present in prokaryotes and in chloroplasts. The protein is structurally and functionally similar to tubulin proteins of eukaryotic cells. Which of the following is a likely conclusion to draw from this information?

(A) FtsZ and tubulin proteins were both present in a common ancestor.

(B) Microtubules are involved in the mechanics of photosynthesis.

(C) Tubulin genes are evolutionarily derived from the gene that codes for the FtsZ protein. ✓

(D) The sequences of the genes encoding the FtsZ and tubulin proteins are identical.

Answer C

This option is correct. The evidence states that the FtsZ gene is present in prokaryotes and in chloroplasts. Chloroplasts evolved from endosymbiotic prokaryotes. The tubulin protein is structurally and functionally similar to FtsZ and is found in eukaryotes, inferring evolutionary descent.

3 Cellular Energetics MCQ

93. The O_2 released during photosynthesis comes from

(A) CO_2

(B) H_2O ✓

(C) NADPH

(D) RuBP (RuDP)

(E) $C_6H_{12}O_6$

94.

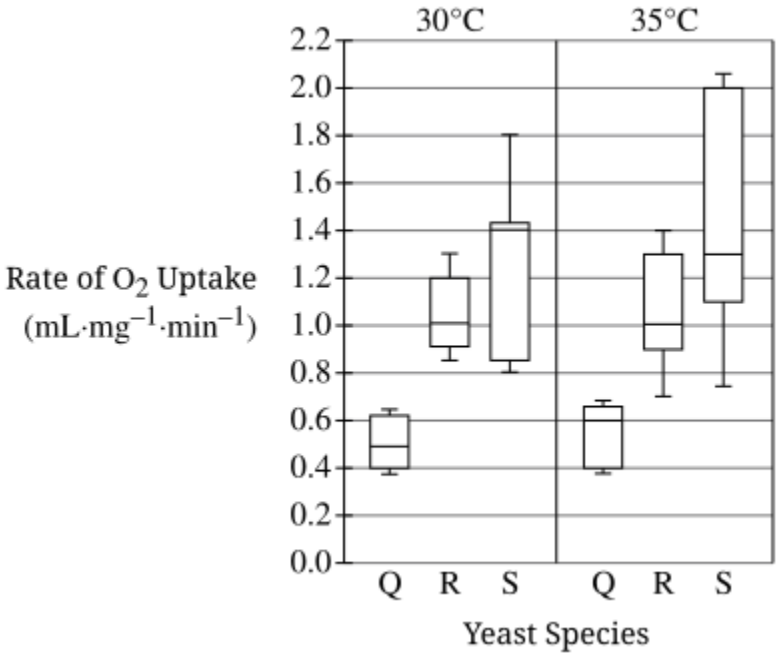


Figure 1. The rate of O_2 uptake for three species of *Saccharomyces* yeast at $30^\circ C$ and $35^\circ C$

Researchers studied the effect of increased temperature on the rate of respiration of three species of *Saccharomyces* yeast (Q, R, and S) by measuring the rate of oxygen uptake in each species at $30^\circ C$ and at $35^\circ C$ (Figure 1).

Based on Figure 1, which of the following statements about the effect of the temperature increase on the respiratory rate of one of the species, Q, R, or S, is most likely true?

(A) The median O_2 uptake rate for species S at $35^\circ C$ is $0.1\ \mu L \cdot mg^{-1} \cdot min^{-1}$ less than it is at $30^\circ C$. ✓

(B) The range from minimum to maximum O_2 uptake rate for species R at $35^\circ C$ is $0.1\ \mu L \cdot mg^{-1} \cdot min^{-1}$ more than it is at $30^\circ C$.

(C) The median O_2 uptake rate for species Q at $35^\circ C$ is the same as it is at $30^\circ C$.

(D) The range of the O_2 uptake rate for species S at $35^\circ C$ is the same as it is at $30^\circ C$.

3 Cellular Energetics MCQ

Answer A

Correct. In a box-and-whisker plot, the line inside the box is the median value. For species S at $30\,^{\circ}C$, the median O_2 uptake rate is 1.4, while at $35\,^{\circ}C$, the median O_2 uptake rate is 1.3.

95. Acetylcholinesterase $(AChE)$ is a protein that catalyzes the conversion of acetylcholine to acetate and choline. When the concentration of $AChE$ in an aqueous solution is held constant, the rate of the reaction catalyzed by $AChE$ increases with increasing concentrations of substrate. At low concentrations of acetylcholine, a small increase in the substrate concentration results in a large increase in the reaction rate. At high concentrations of acetylcholine, however, a large increase in the substrate concentration results in only a small increase in the reaction rate.

Which of the following statements correctly explains the observed effect of the acetylcholine concentration on the rate of the enzyme-catalyzed reaction?

(A) The active site of $AChE$ is specific for acetylcholine, and only one substrate molecule can occupy the active site at a time. ✓

(B) $AChE$ begins converting product into substrate as the acetylcholine concentration changes from low to high.

(C) The $AChE$ protein becomes denatured as the acetylcholine concentration changes from low to high.

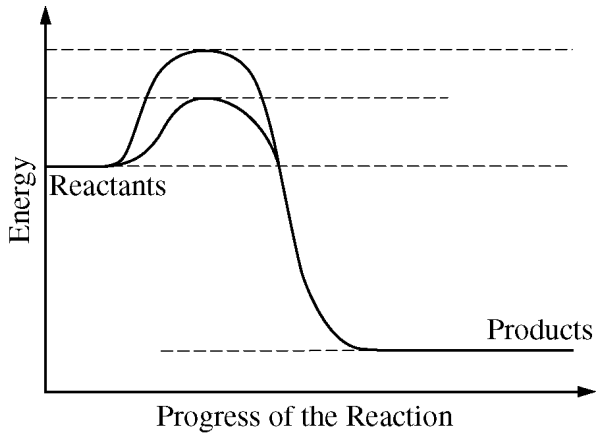
(D) The substrate specificity of $AChE$ changes as the acetylcholine concentration changes from low to high.

Answer A

Correct. Acetylcholine is the substrate for $AChE$, and since the shape and charge of the active site is specific for acetylcholine, only one substrate molecule can occupy the active site at a time. Therefore, at high concentrations of acetylcholine, $AChE$ becomes saturated with substrate. When an enzyme is saturated with substrate, adding more substrate to the solution will not lead to an observable increase in the reaction rate.

3 Cellular Energetics MCQ

96. The diagram below shows energy changes in a specific chemical reaction with and without the addition of an enzyme to the reaction.



Which of the following questions can best be answered by the diagram?

(A) Does the addition of an enzyme reduce the activation energy required for a reaction? ✓

(B) Does the addition of an enzyme result in the formation of covalent bonds?

(C) Does the addition of an enzyme produce a greater amount of products?

(D) Does the addition of an enzyme change the pathway for the reaction?

97. A researcher proposes a model of an enzyme-catalyzed reaction in which a reactant is converted to a product. The model is based on the idea that the reactant passes through a transition state within the enzyme-substrate complex before the reactant is converted to the product.

Which of the following statements best helps explain how the enzyme speeds up the reaction?

(A) The enzyme's active site binds to and stabilizes the reactant, which decreases the free-energy change of the reaction.

(B) The enzyme's active site binds to and stabilizes the transition state, which decreases the activation energy of the reaction. ✓

(C) The enzyme's active site binds to and stabilizes the product, which increases the amount of energy released by the reaction.

(D) The enzyme's active site binds to and stabilizes both the reactant and the product at the same time, which increases the reaction's equilibrium constant.

Answer B

Correct. An enzyme typically speeds up a reaction by stabilizing a transition state in the reaction, which decreases the reaction's activation energy. The enzyme's active site binds more tightly to the transition state than to either the substrate or the product, which speeds up the reaction and allows the enzyme to

catalyze the same reaction repeatedly.

98. Amylase is a protein that catalyzes the conversion of starch to simple sugars. Amylase activity in an aqueous solution can be measured by using iodine as a starch indicator. A solution containing iodine and starch will have a dark-blue color, whereas a solution containing iodine but no starch will have a light-brown color. The color change of an iodine solution from dark blue to light brown can be used to measure the rate at which starch is converted to simple sugars.

A student designs an experiment to investigate the effect of environmental pH on amylase function. The design of the experiment is presented in Table 1.

Table 1. An experiment for investigating the effect of pH on amylase function

Test Tube	Environmental pH	Starch Added	Amylase Added	Iodine Added	Pretreated by Boiling
I	5	Yes	Yes	Yes	No
II	6	Yes	Yes	Yes	No
III	7	No	Yes	Yes	No
IV	7	Yes	No	Yes	No
V	7	Yes	Yes	Yes	Yes
VI	7	Yes	Yes	Yes	No
VII	8	Yes	Yes	Yes	No
VIII	9	Yes	Yes	Yes	No

Which of the following statements best justifies the inclusion of test tube V as a control in the experiment?

(A) It will provide a measurement of amylase activity at an acidic pH.

(B) It will provide a measurement of amylase activity at a basic pH.

(C) It will show the color change that occurs in the absence of enzyme activity. ✓

(D) It will show the color change that occurs in the absence of the amylase protein.

Answer C

Correct. Test tube V will contain all the components of the reaction mixture, but the amylase protein will be inactive because it will have been denatured during the pretreatment. Test tube V is an appropriate control in the experiment because it will show the color change that occurs in the absence of amylase activity.

3 Cellular Energetics MCQ

99. A researcher claims that some bacteria contain factors that influence the function of a particular enzyme but other bacteria do not. To test the claim experimentally, the researcher will grow two different bacterial strains in separate liquid cultures and isolate the contents of the cells in each culture. The researcher will add different combinations of cellular contents, substrate, and enzyme to test tubes containing a buffer solution adjusted to the optimal pH of the enzyme and then measure the rate of product formation. The design of the researcher's experiment is presented in Table 1.

Table 1. An experiment for investigating the influence of bacterial factors on enzyme function

Test Tube	Source of Cellular Contents	Cellular Contents Added	Substrate Added	Enzyme Added
1	Strain 1	No	Yes	Yes
2	Strain 1	Yes	No	Yes
3	Strain 1	Yes	Yes	No
4	Strain 1	Yes	Yes	Yes
5	Strain 2	No	Yes	Yes
6	Strain 2	Yes	No	Yes
7	Strain 2	Yes	Yes	No
8	Strain 2	Yes	Yes	Yes

Which of the following statements best justifies the inclusion of test tubes 3 and 7 in the experiment?

(A) They will show whether the isolated cellular contents have enzymatic activity. ✓

(B) They will show whether environmental pH affects the function of the enzyme.

(C) They will show the rate of product formation in the absence of bacterial factors.

(D) They will show the rate of product formation in the absence of the substrate.

Answer A

Correct. The researcher will add cellular contents and the substrate but not the enzyme to test tubes 3 and 7. As such, test tubes 3 and 7 will show whether the cellular contents have enzymatic activity.

100. A researcher designs an experiment to investigate the effect of environmental temperature on the function of an enzyme. For each trial included in the experiment, the researcher will add the enzyme and its substrate to an aqueous buffer solution and then measure the amount of product formed over 20 minutes.

Which of the following must remain the same for all trials of this experiment?

3 Cellular Energetics MCQ

(A) The initial concentration of the substrate ✓

(B) The final concentration of the product

(C) The three-dimensional structure of the enzyme

(D) The temperature of the aqueous buffer solution

Answer A

Correct. Keeping the initial substrate concentration constant allows the researcher to compare the reaction rates under different environmental conditions. The reaction rates will provide the researcher with information about the enzyme's function in different environments.

101. Catalase is an enzyme found in yeast cells that facilitates the chemical breakdown of hydrogen peroxide to water and oxygen gas. An experiment was conducted to determine the effect of pH on catalase function. Five buffer solutions of varying pH (2, 4, 6, 8, and 10) were prepared and added to separate test tubes. Hydrogen peroxide (H_2O_2) was added to each test tube. Yeast was added, and the reactions were timed. After 1 minute the amount of oxygen gas released was determined by measuring the foam layer produced in each test tube. Figure 1 illustrates the experimental setup.

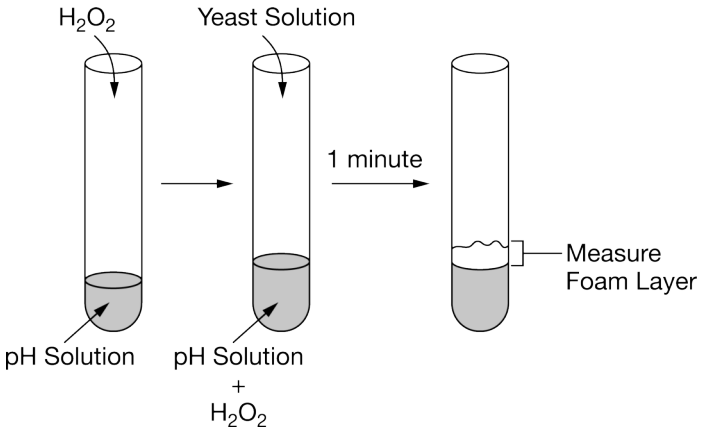


Figure 1. Illustration of experimental procedure

A set of five additional test tubes were prepared and used as controls. Which of the following best describes the contents expected to be contained in one of the five control test tubes?

(A) pH 4 buffer solution and hydrogen peroxide only ✓

(B) All five pH buffer solutions combined and hydrogen peroxide only

(C) Water, hydrogen peroxide, and yeast

(D) Water and yeast only

3 Cellular Energetics MCQ

Answer A

Correct. Since the experiment is studying the effect of pH on catalase in yeast, the control set should determine what would happen without yeast present. Each test tube in the control set should contain hydrogen peroxide and one of the pH solutions.

For following group of questions first study the description of the data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

To study the actions of the enzyme catalase on hydrogen peroxide, students performed the following experiment. Catalase was extracted from potatoes by blending raw potatoes in a blender with cold distilled water. The filtrate was stored on ice. The following hydrogen peroxide solutions were made: 1 percent, 5 percent, 10 percent, and 15 percent. Filter paper disks were soaked in the catalase filtrate and dropped into beakers containing the various solutions. The activity of the enzyme was measured by the amount of time it took for the disks to float to the surface of the solution on the bubbles produced by the reaction. The following data were obtained.

Hydrogen Peroxide Solution	Average Time, in Seconds, for Disks to Float
1%	30 sec
5%	25 sec
10%	20 sec
15%	10 sec

102. Which of the following experimental designs should the students use as a control for the experiment?

(A) Place a catalase-soaked disk in a beaker of water. ✓

(B) Drop the disks from different heights into the solution.

(C) Poke the floating disks back down to the bottom of the beaker and retime the rise of the disks.

(D) Shake the beakers during the time the disks are rising.

(E) Dip the disks in the 5% solution before putting them in any other solution.

103. Which of the following best describes why ice was used during this experiment?

(A) To increase the activity of the enzyme

(B) To retard the breakdown of the catalase ✓

(C) To purge the solution of excess air trapped during blending

(D) To slow the catalase molecules to increase the probability of contact with the hydrogen peroxide molecules

(E) To increase the size of the active site on the enzyme

3 Cellular Energetics MCQ

104. Which of the following best explains why the disks rose to the surface faster in the more concentrated hydrogen peroxide solutions?

(A) There was more enzyme present in the more concentrated solutions.

(B) A greater amount of heat was generated in the more concentrated solutions.

(C) The more concentrated solutions lowered the activation energy of the reaction.

(D) The higher substrate concentrations in the more concentrated solutions speeded the reaction. ✓

(E) The density of the water was lower in the more concentrated solutions.

105. If the potato solution was boiled for 10 minutes and cooled for 10 minutes before being tested, the average time for the disks to float to the surface of the hydrogen peroxide solution would be

(A) less than 1 second

(B) 5 seconds

(C) 10 seconds

(D) 30 seconds

(E) more than 30 seconds ✓

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106. Which of the following statements best helps explain the reaction specificity of an enzyme?

(A) The free energy of the reactants is greater than the free energy of the products.

(B) The equilibrium constant of the reaction is much greater than 1.

(C) The shape and charge of the substrates are compatible with the active site of the enzyme. ✓

(D) The concentration of the enzyme inside living cells is greater than the concentration of substrate.

Answer C

Correct. The reaction specificity of an enzyme is determined by the compatibility of the substrate with the enzyme's active site. A substrate's shape and charge affect how it fits into an enzyme's active site and whether it forms favorable interactions with the enzyme.

3 Cellular Energetics MCQ

107. Gelatin is a protein that is derived from collagen which is found in the bones, skin, and connective tissue of animals. To investigate the ability of various enzymes to digest gelatin, a group of students set up an assay involving camera film. Camera film contains gelatin and appears black when exposed to light but turns clear as the gelatin gets broken down. The students incubated pieces of exposed camera film in test tubes, each containing one of three different enzyme solutions (trypsin, lipase, or amylase) as indicated in Figure 1. The students recorded the time it took for the enzymes to digest the gelatin in each test tube, turning the film from black to clear.

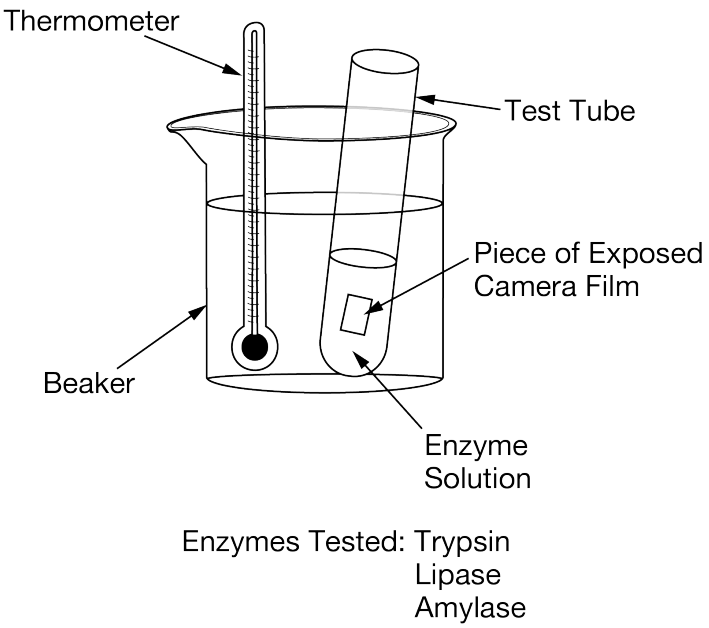


Figure 1. Diagram of experimental setup.

Which of the following would be the most appropriate control for this experiment?

(A) A test tube containing no camera film

(B) A test tube containing only a piece of exposed camera film

(C) A test tube containing a piece of exposed camera film submerged in water ✓

(D) A test tube containing a piece of exposed camera film and all three enzyme solutions

Answer C

Correct. Having one test tube with only water and camera film will show that water is not the substance that causes the film to change color.

3 Cellular Energetics MCQ

108. The enzyme hexokinase catalyzes the conversion of glucose to glucose-6-phosphate, which is an important step in glycolysis. The reaction involves the transfer of a phosphate group from ATP to glucose.

Either a glucose molecule or a water molecule can fit in the active site of hexokinase. The presence of a water molecule in hexokinase's active site would result in the hydrolysis of ATP to ADP instead of the conversion of glucose to glucose-6-phosphate.

Which of the following statements best helps explain the reaction specificity of hexokinase?

(A) Both glucose and water are polar molecules that form favorable interactions with charged and polar amino acid side chains in hexokinase's active site.

(B) Both glucose and water have oxygen atoms that can form covalent bonds with the phosphorus atoms of phosphate groups.

(C) Glucose is an energy-rich organic molecule that can be broken down by glycolysis to produce ATP, whereas water is an inorganic molecule.

(D) Glucose has the right shape and charge to cause hexokinase to undergo a structural change needed for catalysis, whereas water does not. ✓

Answer D

Correct. Hexokinase catalyzes the conversion of glucose to glucose-6-phosphate but not the hydrolysis of ATP. Hexokinase's reaction specificity is explained by the compatibility of glucose and hexokinase's active site, which results in part from glucose's shape and charge. Because of this compatibility, hexokinase undergoes a structural change when it binds to glucose but not to water, and the structural change is necessary for hexokinase to become catalytically active.

3 Cellular Energetics MCQ

109. A student designs an experiment to investigate the influence of temperature on enzyme function. The student's plan is presented in Table 1.

Table 1. An experiment to investigate the influence of temperature on enzyme function

Test Tube	Substrate Added	Enzyme Added	Temperature (°C)
1	Yes	No	10
2	Yes	Yes	10
3	Yes	No	20
4	Yes	Yes	20
5	Yes	No	30
6	Yes	Yes	30
7	Yes	No	40
8	Yes	Yes	40

Which test tubes are controls in the experiment?

(A) Test tubes 1 and 2 only

(B) Test tubes 5 and 6 only

(C) Test tubes 1, 3, 5, and 7 ✓

(D) Test tubes 2, 4, 6, and 8

Answer C

Correct. The test tubes that do not contain the enzyme are negative controls in the experiment. The negative controls help confirm that the results of the experiment are related to enzyme function and not some other factor.

110. Aminolevulinate dehydratase (ALAD) is an enzyme that relies on zinc as a cofactor. A zinc ion binds to the ALAD active site, where it forms favorable interactions with the side chains of three amino acids. Researchers have found that substituting a lead ion for a zinc ion in the ALAD active site causes inhibition of ALAD.

Which of the following statements best helps explain how the lead ion causes inhibition of ALAD?

3 Cellular Energetics MCQ

(A) It changes the shape and charge of the substrate so that it becomes more compatible with ALAD's active site.

(B) It changes the amino acid sequence of the ALAD protein so that the enzyme catalyzes a different reaction.

(C) It changes the three-dimensional structure of the active site so that ALAD is no longer compatible with its substrate. ✓

(D) It changes the enzyme-substrate complex so that the transition state is more stable and the reaction proceeds at a faster rate.

Answer C

Correct. The substitution of a lead ion for a zinc ion most likely changes the three-dimensional structure of the ALAD active site. The three-dimensional structure of the active site plays an important role in the compatibility between an enzyme and its substrate. Changing the structure of the active site will likely interfere with enzyme function by disrupting the enzyme-substrate interaction.

111. Pectinase is a protein that catalyzes the breakdown of pectic polysaccharides in plant cell walls. A researcher designs an experiment to investigate the effect of salinity on the ability of pectinase to lower the activation energy of the reaction involved. The design of the experiment is presented in Table 1. For each test tube, the researcher will measure the amount of product formed over 20 minutes.

Table 1. An experiment to investigate the effect of salinity on pectinase function

Test Tube	Sodium Chloride Concentration (molar)	Temperature (degrees C)	Substrate Added	Pectinase Added
1	0	23	Yes	No
2	0	23	Yes	Yes
3	0.5	23	Yes	No
4	0.5	23	Yes	Yes
5	1.0	23	Yes	No
6	1.0	23	Yes	Yes
7	1.5	23	Yes	No
8	1.5	23	Yes	Yes

Which of the following statements best helps justify the inclusion of test tube 5 in the experiment?

3 Cellular Energetics MCQ

(A) It will act as a control for test tube 4 by showing the effect of the presence or absence of the substrate.

(B) It will act as a control for test tube 4 by showing the effect of a change in environmental temperature.

(C) It will act as a control for test tube 6 by showing the effect of the presence or absence of the enzyme. ✓

(D) It will act as a control for test tube 6 by showing the effect of a change in sodium chloride concentration.

Answer C

Correct. Test tube 5 will act as a control for test tube 6. The solutions in test tubes 5 and 6 are nearly identical except that the researcher will add pectinase to test tube 6 but not to test tube 5. A comparison of the data for test tubes 5 and 6 will show the effect of the presence or absence of the enzyme.

Researchers investigated the influence of environmental pH on the activity of peroxidase, an enzyme that catalyzes the conversion of hydrogen peroxide to water and oxygen gas. In an experiment, the researchers added a hydrogen peroxide solution containing guaiacol to several identical test tubes and adjusted the solution in each test tube to a different pH. The researchers included the guaiacol because it caused the solutions to change color as the reactions proceeded, which the researchers relied on for measuring reaction rates. Finally, the researchers added the same amount of peroxidase to each test tube and measured the rate of each reaction at $23^{\circ}C$. The results of the experiment are represented in Figure 1.

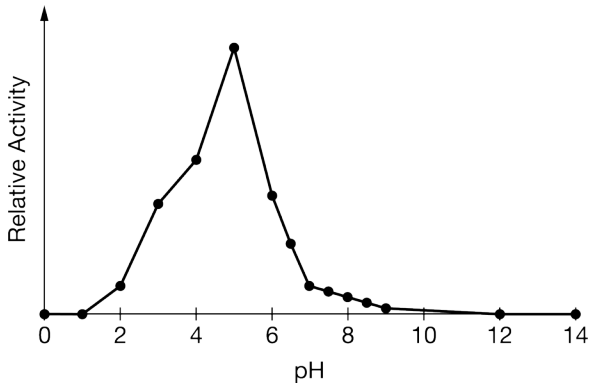


Figure 1. The effect of pH on peroxidase activity

112. One of the researchers proposes using oxygen gas production to measure reaction rates. Which of the following statements best justifies the use of the proposed modification as a way of creating an appropriate control for the investigation?

3 Cellular Energetics MCQ

(A) The experiment can be repeated without hydrogen peroxide, which will help eliminate an uncontrolled variable.

(B) The experiment can be repeated without peroxidase, which will introduce a second independent variable.

(C) The experiment can be repeated without guaiacol, which will reveal the effect of guaiacol on the reaction rates. ✓

(D) The experiment can be repeated without water, which will reveal whether the reaction can occur inside a living cell.

Answer C

Correct. In the original experiment, the researchers used guaiacol as a reaction indicator. By using oxygen gas production to measure reaction rates, the researchers no longer need the guaiacol and can repeat the experiment without it, which will reveal the effect of guaiacol on the reaction rates.

113. Which of the following actions will provide a negative control for the investigation?

(A) Repeating the experiment at $25^\circ C$

(B) Repeating the experiment using twice the amount of hydrogen peroxide

(C) Repeating the experiment using twice the amount of peroxidase

(D) Repeating the experiment using heat-denatured peroxidase ✓

Answer D

Correct. A negative control is a group in which no response is expected. It is the opposite of the positive control, in which a known response is expected. Heat-denatured peroxidase will be inactive. Repeating the experiment using heat-denatured peroxidase will allow the researcher to measure reaction rates in the presence of inactive peroxidase.

114. Based on Figure 1, which of the following statements best predicts the effect that a change from a moderately acidic environment (pH near 6) to a basic environment will have on peroxidase activity?

(A) Peroxidase activity will decrease. ✓

(B) Peroxidase activity will increase.

(C) Peroxidase activity will stay the same.

(D) Peroxidase activity will increase at first and then decrease.

3 Cellular Energetics MCQ

Answer A

Correct. Based on Figure 1, peroxidase's optimal pH is 5. A change from a moderately acidic environment (pH near 6) to a basic environment (pH above 7) will result in a decrease in peroxidase activity.

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115. A researcher designs an experiment to investigate whether soil bacteria trigger the synthesis of defense enzymes in plant roots. The design of the experiment is presented in Table 1. For each group in the experiment, the researcher will determine the average rate of change in the amount of defense enzymes in the roots of the seedlings.

Table 1. An experiment to investigate the effect of soil bacteria on plant defenses

Group	Number of Seedlings	Type of Soil	Treatment Solution
1	10	Sterile potting soil	Contains actively reproducing soil bacteria
2	10	Sterile potting soil	Contains heat-killed soil bacteria
3	10	Sterile potting soil	Contains no soil bacteria

Which of the following statements best helps justify the inclusion of group 2 as one of the controls in the experiment?

(A) It will show whether the changes observed in group 1 depend on the metabolic activity of soil bacteria. ✓

(B) It will show whether the changes observed in group 1 depend on the type of plants used in the experiment.

(C) It will show the average growth rate of seedlings that are maintained in a nonsterile environment.

(D) It will show the changes that occur in the roots of seedlings following an infection by soil bacteria.

Answer A

Correct. The seedlings in group 1 will be treated with a solution that contains actively reproducing bacteria, which will be metabolically active. In contrast, the seedlings in group 2 will be treated with a solution that contains heat-killed bacteria, which will be metabolically inactive. The researcher will be able to compare the results for group 1 with those for group 2. The comparison will help the researcher determine whether the changes observed in group 1 depend on the metabolic activity of soil bacteria.

3 Cellular Energetics MCQ

116. A researcher claims that increasing the environmental temperature by $10^{\circ}C$ will double the rate of an enzyme-catalyzed reaction. To test the claim, the researcher designs an experiment that uses a particular enzyme isolated from plants. The design of the experiment is presented in Table 1. For each test tube in the experiment, the researcher will measure the rate of product formation.

Test Tube	Temperature ($^{\circ}C$)	Substrate Added	Enzyme Added	Pretreated by Boiling
1	15	Yes	Yes	No
2	25	No	Yes	No
3	25	Yes	No	No
4	25	Yes	Yes	No
5	25	Yes	Yes	Yes
6	35	Yes	Yes	No

Which of the following statements best helps justify the inclusion of test tube 5 as a control in the experiment?

(A) It will provide a measurement of product formation in the absence of the substrate.

(B) It will provide a measurement of product formation in the presence of a denatured enzyme. ✓

(C) It will show the effect of doubling the amount of substrate on the rate of product formation.

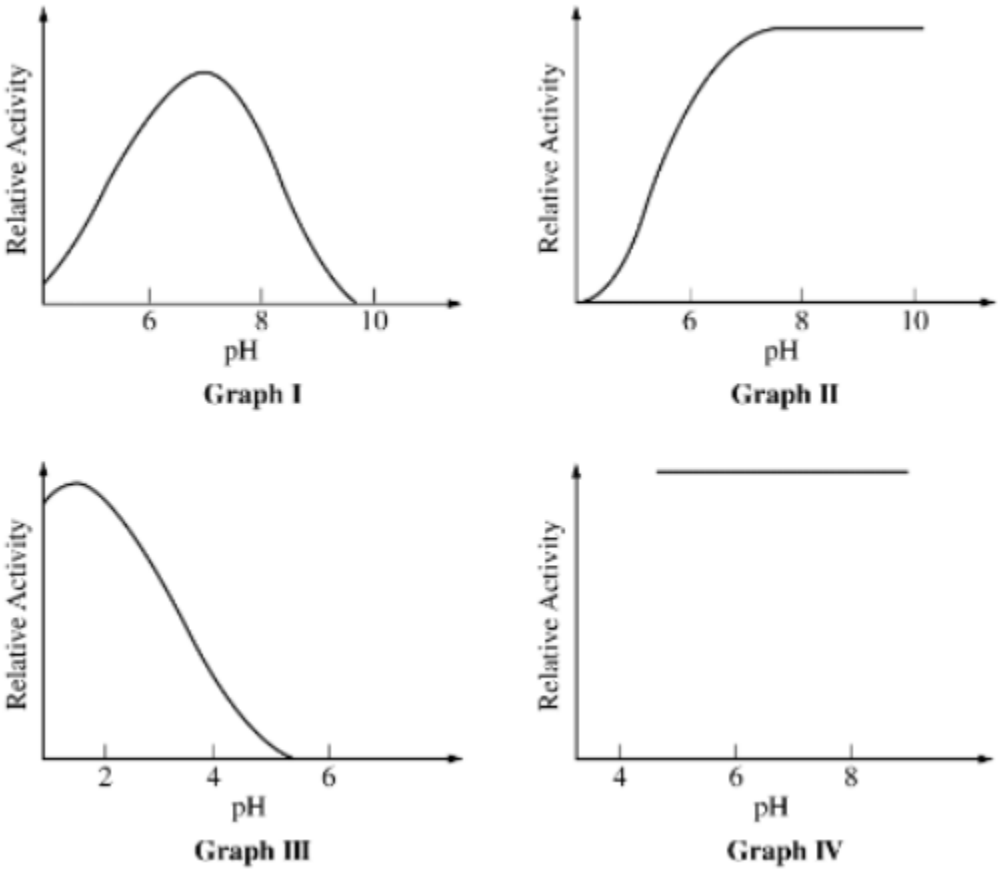
(D) It will show the effect of increased enzyme activity on the rate of product formation.

Answer B

Correct. The contents of test tube 5 will be pretreated by boiling. The pretreatment will denature the plant enzyme used in the experiment. As such, test tube 5 will provide a measurement of product formation in the presence of a denatured enzyme.

3 Cellular Energetics MCQ

For the following questions: Graphs I-IV depict the effect of pH on the activity of four different hydrolytic enzymes.



117. Graphs representing enzymes sensitive to changes in pH include which of the following?

(A) I only
(B) IV only
(C) II and III only
(D) III and IV only
(E) I, II, and III only ✓

118. The most likely explanation for the results shown in Graph I is that

(A) pH affects the shape of the active site of the enzyme ✓
(B) pH affects the temperature of the reaction
(C) the enzyme has a quaternary structure
(D) the enzyme has disulfide bonds
(E) pH affects the primary structure of the enzyme

119. Enzymes with their highest activity at an alkaline (basic) pH are represented by which of the following graphs?

3 Cellular Energetics MCQ

(A) I only

(B) II only ✓

(C) III only

(D) I and III only

(E) I and IV only

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120. In an experiment, a researcher prepares a reaction mixture by dissolving a substance in a buffered solution. The substance is the substrate of a certain enzyme. The researcher adds a small amount of the enzyme to the reaction mixture and measures the amount of product that is formed over time. The data are represented in Figure 1.

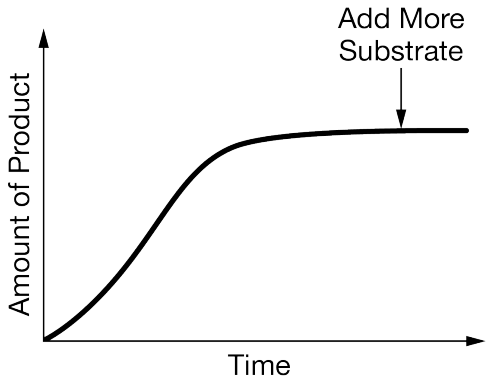


Figure 1. The amount of product formed by an enzyme-catalyzed reaction over time

Which of the following best predicts the immediate result of adding more substrate to the reaction mixture at the point indicated by the arrow in Figure 1?

(A) The amount of product will decrease until the reaction rate goes to zero.

(B) The amount of product will decrease until the reaction reaches its equilibrium point or until the enzyme is been used up by the reaction.

(C) The amount of product will increase until the reaction reaches its equilibrium point or until the substrate is used up by the reaction. ✓

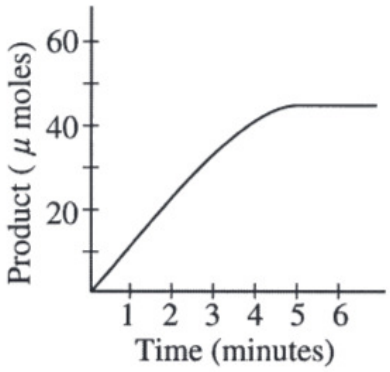
(D) The amount of product will increase without stopping because the enzyme will be unchanged by the reaction.

Answer C

Correct. The graph shown in Figure 1 indicates that the amount of product became constant as a consequence of the reaction reaching its equilibrium point or as a consequence of the substrate being used up. Adding more substrate will likely result in an increase in the amount of product because the reaction catalyzed by the enzyme will convert the added substrate into product.

3 Cellular Energetics MCQ

121. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



During which time interval is the reaction rate lowest?

(A) 0-1 minute
(B) 1-2 minutes
(C) 2-3 minutes
(D) 3-4 minutes
(E) 4-5 minutes ✓

3 Cellular Energetics MCQ

The coral species *Acropora millepora* is a marine animal that has a symbiotic relationship with algae species in the genus *Symbiodinium*. An increase in the concentration of dissolved CO_2 in oceans has a negative impact on this relationship and the health of the coral species. Scientists investigated the effect of CO_2 concentrations on the number of algae cells found in the coral (Figure 1) and on both the gross (total) and net rates of photosynthesis of those algae (Figure 2). The net rate of photosynthesis is equal to the gross rate of photosynthesis minus the rate of respiration by the algae.

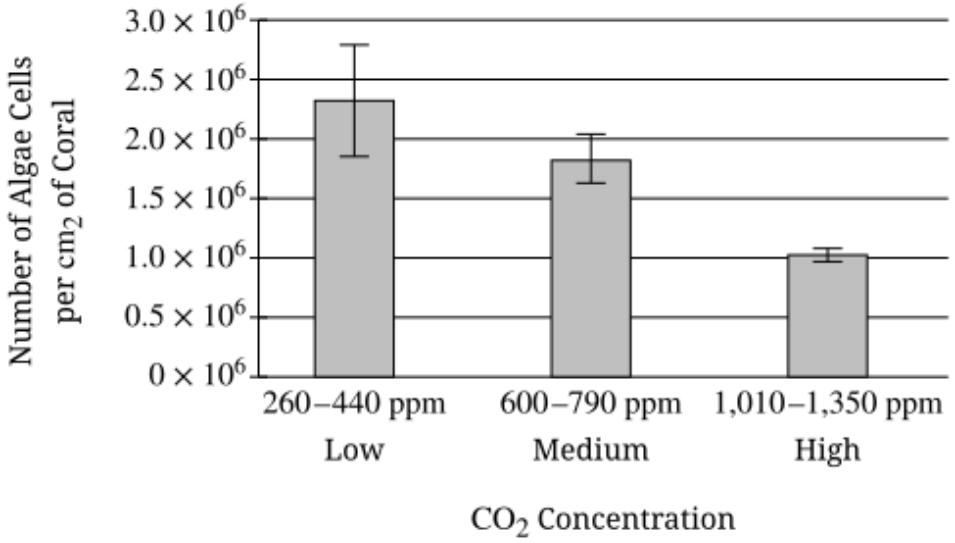


Figure 1. Effect of CO_2 concentration on the number of algae per cm^2 of coral. Error bars represent $\pm 2SE_{\bar{x}}$.

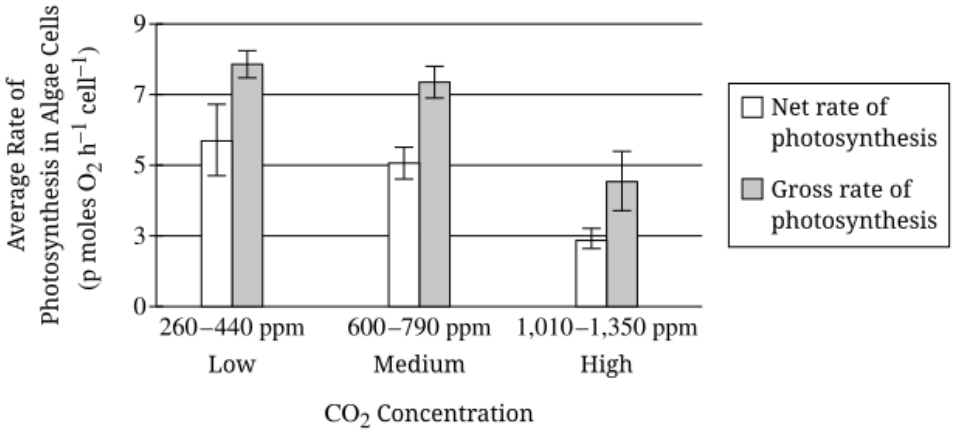


Figure 2. Average gross and net rates of photosynthesis in algae under various concentrations of CO_2. Error bars represent $\pm 2SE_{\bar{x}}$.

3 Cellular Energetics MCQ

122. Researchers observe that the water with the high CO_2 concentration has a lower pH than does the water with the low CO_2 concentration. The researchers claim that lower pH alters the structure of the enzymes of photosynthesis.

Which of the following most likely explains how a lower pH could alter the structure of photosynthetic enzymes?

(A) There will not be enough H^+ ions for the ribosome to correctly form peptide bonds between amino acids during synthesis of the enzymes.

(B) There will not be enough H^+ ions for the necessary hydrogen bonds to form in the secondary, tertiary, and quaternary structures of the enzymes.

(C) There will be additional H^+ ions that can disrupt peptide bonds in the primary structure of the enzymes.

(D) There will be additional H^+ ions that can disrupt hydrogen bonds in the secondary, tertiary and quaternary structures of the enzymes. ✓

Answer D

Correct. Additional H^+ ions compete for hydrogen bonding with amino acid side chains and therefore denature the protein and alter the structure of the photosynthetic enzymes.

123. Based on Figure 2, which of the following best supports the claim that CO_2 concentration and photosynthetic productivity are inversely related?

(A) The net rate of photosynthesis is less than the gross rate of photosynthesis at the high concentration of CO_2.

(B) Both the net and gross rates of photosynthesis at the high concentration of CO_2 are lower than those at the low concentration of CO_2. ✓

(C) The net rate of photosynthesis at the low concentration of CO_2 is lower than the gross rate of photosynthesis at the high concentration of CO_2.

(D) The difference between net and gross rates of photosynthesis is greater at the low concentration of CO_2 than it is at the high concentration of CO_2.

Answer B

Correct. This describes an inverse relationship where one measure increases as the other decreases. Because the rate of photosynthesis decreases as CO_2 concentration increases, this supports the claim that CO_2 concentration is inversely related to photosynthetic rate (productivity).

3 Cellular Energetics MCQ

124. Alcohol dehydrogenase (ADH) is an enzyme that aids in the decomposition of ethyl alcohol ($\mathrm{C_2H_5OH}$) into nontoxic substances. Methyl alcohol acts as a competitive inhibitor of ethyl alcohol by competing for the same active site on ADH. When attached to ADH, methyl alcohol is converted to formaldehyde, which is toxic in the body.

Which of the following statements best predicts the effect of increasing the concentration of substrate (ethyl alcohol), while keeping the concentration of the inhibitor (methyl alcohol) constant?

(A) There will be an increase in formaldehyde because ADH activity increases.

(B) Competitive inhibition will be terminated because ethyl alcohol will bind to methyl alcohol and decrease ADH activity.

(C) The peptide bonds in the active site of the enzyme will be denatured, inhibiting the enzyme.

(D) Competitive inhibition will decrease because the proportion of the active sites occupied by substrate will increase. ✓

Answer D

Correct. The increase in amount of substrate will increase the number of enzyme-ethanol complexes formed, increasing the amount of nontoxic product formed from ethyl alcohol.

125. A researcher proposes a model to explain how enzyme-substrate interactions determine enzyme specificity. The model is based on the idea that substrate molecules form favorable interactions with the amino acid side chains in an enzyme's active site.

Based on the model, which of the following statements best explains an enzyme's specificity for a particular substrate molecule?

(A) A hydrophilic molecule interacts with nonpolar side chains in the enzyme's active site.

(B) A hydrophobic molecule interacts with polar side chains in the enzyme's active site.

(C) A molecule with positive charges interacts with positively charged side chains in the enzyme's active site.

(D) A molecule with negative charges interacts with positively charged side chains in the enzyme's active site. ✓

Answer D

Correct. Because opposite charges attract, a molecule with negative charges will form favorable interactions with positively charged side chains in an enzyme's active site. Based on the model, the favorable interactions helps explain the enzyme's specificity for the molecule as a substrate.

3 Cellular Energetics MCQ

126. A researcher claims that different enzymes exhibit maximal function over different pH ranges. To test the claim, the researcher carries out an experiment that includes three different enzymes: pepsin, salivary amylase, and arginase. The results of the experiment are represented in Figure 1.

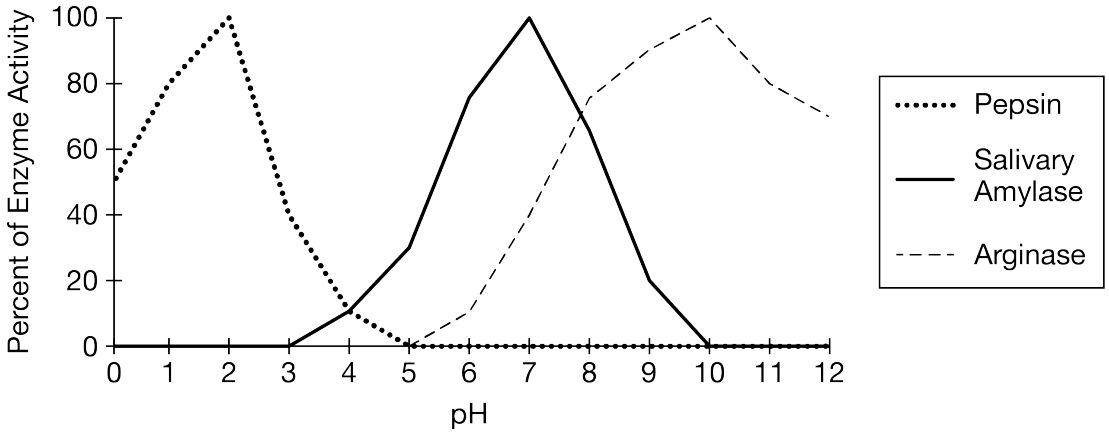


Figure 1. The effect of pH on three different enzymes

Which of the following actions will provide the most appropriate negative control for the experiment?

(A) Repeating the experiment with a fourth enzyme

(B) Repeating the experiment at several different temperatures

(C) Repeating the experiment with denatured enzymes ✓

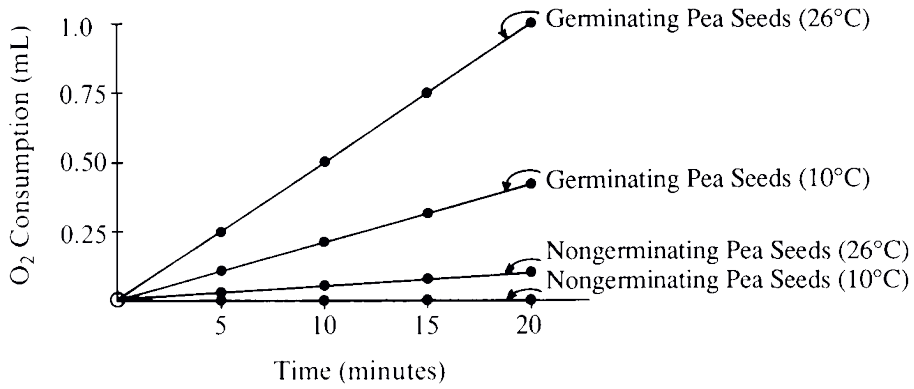
(D) Repeating the experiment using several different methods for measuring pH

Answer C

Correct. Repeating the experiment with denatured enzymes will provide an appropriate negative control for the experiment. The negative control will show that the changes observed in the experiment result from changes in enzyme function.

3 Cellular Energetics MCQ

The following questions refer to an experiment that is set up to determine the relative volume of O_2 consumed by germinating and nongerminating (dry) pea seeds at two different temperatures. The change in volume is detected by using a respirometer over a given period of time. The data are given below.



127. The rate of oxygen consumption in germinating pea seeds at 26ºC is

(A) 0.05 mL / min ✓

(B) 0.25 mL / min

(C) 0.50 mL / min

(D) 0.75 mL / min

(E) 1.00 mL / min

128. Which of the following conclusions is best supported by the data?

(A) Nongerminating pea seeds have a higher rate of respiration than germinating pea seeds do.

(B) Light is required for pea seed germination.

(C) In the nongerminating pea seeds, oxygen consumption is directly proportional to oxygen concentration.

(D) Less carbon dioxide is produced by germinating pea seeds at 26ºC than at 10ºC.

(E) In pea seeds an increase in temperature results in an increase in oxygen consumption. ✓

3 Cellular Energetics MCQ

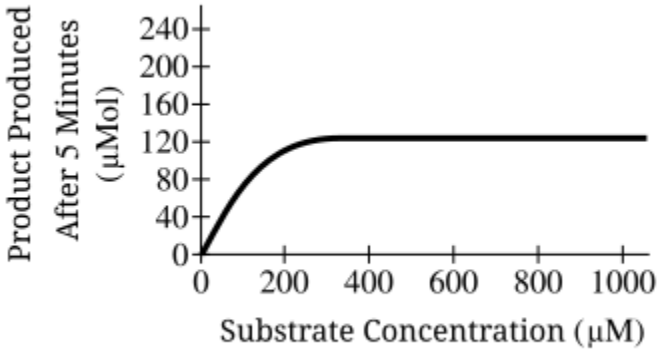


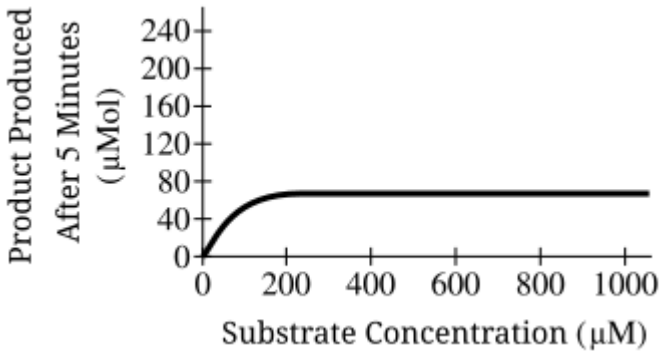
Figure 1. Rate of an enzyme-catalyzed reaction

Researchers investigated the dynamics of a reaction catalyzed by an enzyme. The researchers prepared a series of reactions, each with the same concentration of enzyme but with different concentrations of substrate. The researchers measured the amount of product in each reaction mixture after 5 minutes. The results are shown in Figure 1.

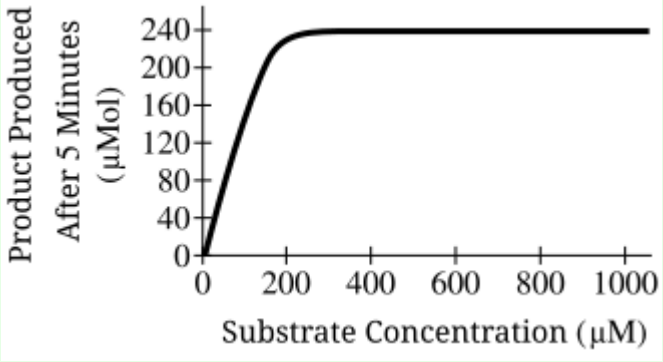
129. Which of the following graphs predicts the most likely rate of the enzyme-catalyzed reaction if the concentration of the enzyme in the mixture were doubled?

3 Cellular Energetics MCQ

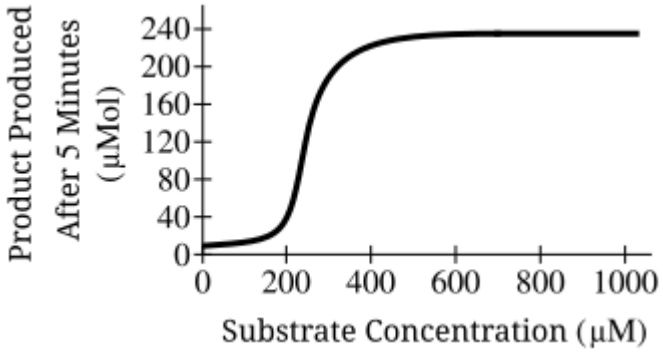
(A)



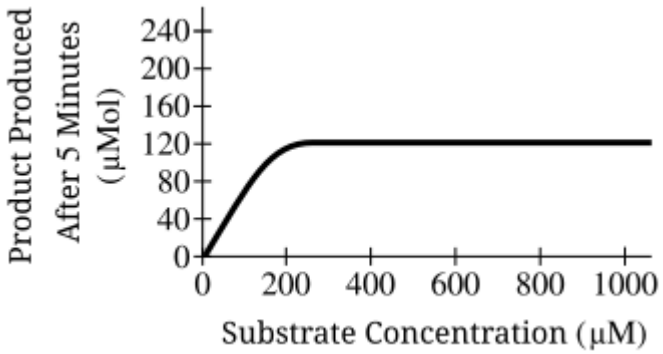
(B)



(C)



(D)



3 Cellular Energetics MCQ

Answer B

Correct. The rate of the reaction is initially proportional to the concentration of enzyme present. When all active sites are occupied, the rate remains constant at a maximum. Thus, doubling the enzyme concentration should double the rate of the reaction at every substrate concentration, and the maximum rate should also double.

130. Which of the following questions best addresses whether a particular inhibitor is competitive or noncompetitive?

(A) Does the inhibitor bind more tightly to the substrate or to the enzyme?

(B) Can the inhibitor be converted to product by the enzyme?

(C) How much inhibitor is necessary to completely inhibit the activity of the enzyme?

(D) Does the inhibitor bind to an allosteric site or the active site of the enzyme? ✓

Answer D

Correct. Competitive enzyme inhibitors affect the rate of reaction by competing with the substrate for binding to the active site of the enzyme. Noncompetitive inhibitors affect the rate of reaction by binding to an allosteric site of the enzyme, altering the shape and thus the function of the enzyme.

131. Which of the following experimental designs would best allow researchers to investigate the effect of pH on the rate of the enzyme-catalyzed reaction?

(A) Measure reaction rates at different pH levels when the enzyme concentration remains the same and the substrate concentration is 100 µM. ✓

(B) Measure the pH of the reaction mixture when the enzyme concentration remains the same and substrate concentrations vary between 0 µM and 1,000 µM.

(C) Measure reaction rates when the enzyme concentration remains the same but the pH of the reaction mixture increases and the substrate concentration increases from 0 µM to 1,000 µM.

(D) Measure reaction rates when the substrate concentration is 100 µM and both the pH of the reaction mixture and the enzyme concentration are increased.

Answer A

Correct. To investigate the role of pH on the reaction rate, the change in pH must be the only independent variable.

3 Cellular Energetics MCQ

132. Which of the following best describes how an increase in the concentration of substrate in the reaction mixture affected the frequency of enzyme-substrate interactions?

(A) The frequency of enzyme-substrate interactions decreased over all substrate concentrations.

(B) The frequency of enzyme-substrate interactions increased exponentially over all substrate concentrations.

(C) The frequency of enzyme-substrate interactions increased until all of the active sites were interacting with substrate. ✓

(D) The frequency of enzyme-substrate interactions increased until all of the enzyme was depleted.

Answer C

Correct. The frequency of enzyme-substrate interactions is determined by the maximum number of enzyme active sites available to bind substrate and convert it to product. The rate of the reaction increases until the active sites of all enzyme molecules are filled, and then the rate of the reaction remains constant.

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133. Which of the following can be used to determine the rate of enzyme-catalyzed reactions?

(A) Rate of disappearance of the enzyme

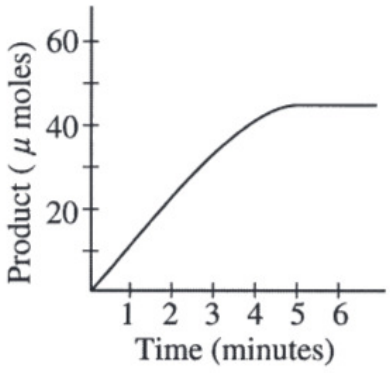
(B) Rate of disappearance of the substrate ✓

(C) Rate of disappearance of the product

(D) Change in volume of the solution

(E) Increase in activation energy

134. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



What is the most likely explanation for the change in the slope of the line between 3 and 5 minutes?

3 Cellular Energetics MCQ

(A) The enzyme had denatured.

(B) The enzyme had achieved its maximum velocity.

(C) A large amount of the substrate had been consumed. ✓

(D) An allosteric inhibitor appeared.

(E) There was a dramatic change in the pH.

135. The enzyme trypsin aids in protein digestion in the small intestine. The relative activity of trypsin at different pH values is shown in Figure 1.

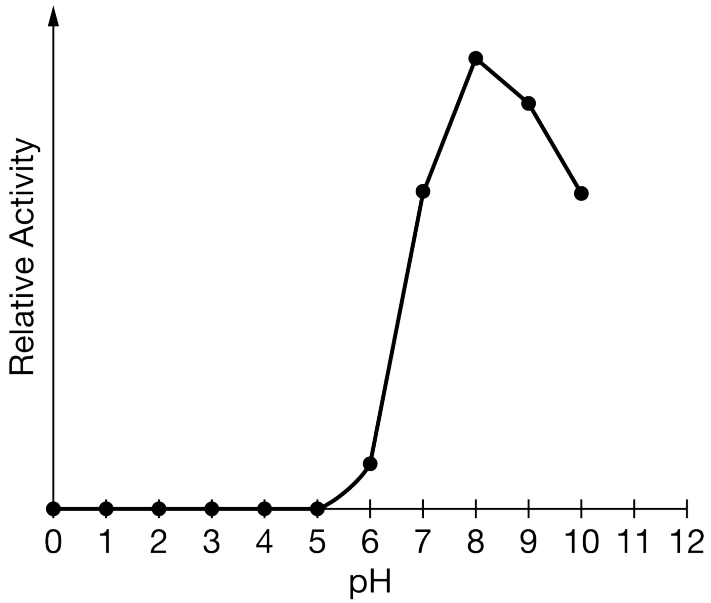


Figure 1. Effect of pH on the activity of trypsin

Which of the following statements best explains the activity levels of trypsin shown in Figure 1 ?

(A) The small intestine releases inhibitor molecules that block the activity of trypsin unless it is at its optimum pH.

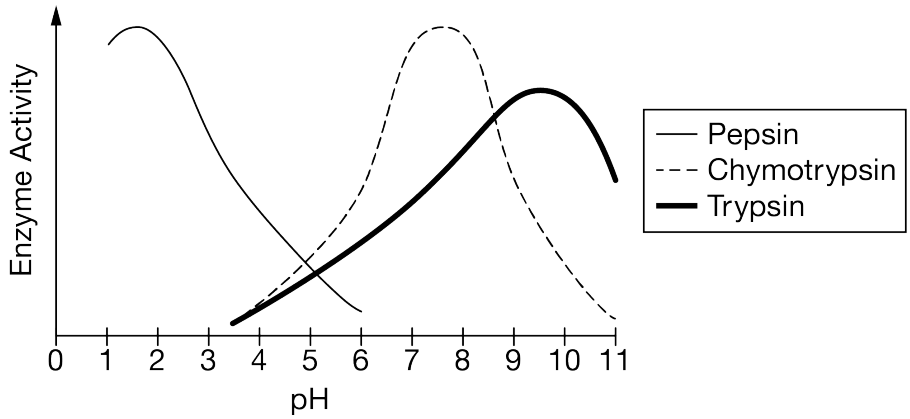
(B) The number of effective collisions between trypsin and its substrate increase at higher pH values.

(C) As pH values increase, the substrate concentration decreases, leading to an eventual decline in the rate of the trypsin-catalyzed reaction.

(D) At extremely low pH values, trypsin is denatured and cannot function efficiently. ✓

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136. Protein digestion in humans is primarily carried out by three enzymes. Pepsin is found in the stomach (pH 2), where it aids in the breakdown of large proteins into smaller peptides, while trypsin and chymotrypsin are found in the small intestine (pH 8), where they aid in the further breakdown of the proteins into amino acids and dipeptides that can be absorbed into the bloodstream. Graph 1 shows the effect of pH on the activity levels of the three enzymes.



Graph 1. Relative activity of pepsin, trypsin, and chymotrypsin at pH 0 through 11

Which of the following best predicts how the structure and function of pepsin will change as it enters the small intestine?

(A) Pepsin will not change in shape and will continue to break down proteins in the small intestine.

(B) Pepsin will not change in shape but may not work due to the basic environment of the small intestine.

(C) Pepsin will change in shape because of the basic environment of the small intestine; therefore, its enzymatic activity will decrease. ✓

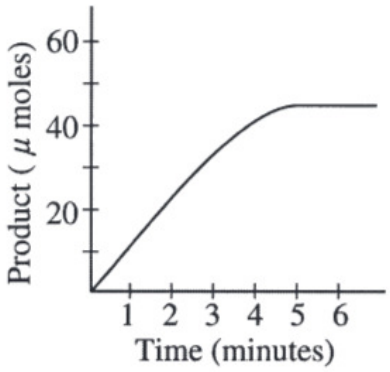
(D) Pepsin will change in shape because of the presence of trypsin and chymotrypsin in the small intestine, both of which act as competitive inhibitors.

Answer C

Correct. The significant change in pH will disturb the hydrogen bonding and cause denaturation of pepsin. As a result, pepsin will stop working in the small intestine.

3 Cellular Energetics MCQ

137. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



The rate of the reaction could also be determined by

(A) measuring the change in the amount of enzyme

(B) measuring the change in the amount of substrate ✓

(C) measuring the change in salt concentration

(D) adding more substrate

(E) adding more enzyme

138. Researchers investigated the effect of urea on the three-dimensional structure of a certain enzyme. The researchers dissolved the enzyme in an aqueous buffer solution and added urea to the solution. The enzyme did not appear to have a secondary or tertiary structure. The researchers carefully removed the urea from the solution and determined that the enzyme had the original secondary and tertiary structure again.

Based on the results of the experiment, which of the following statements best predicts the effect of urea on the enzyme's function?

(A) Function will be disrupted by adding the urea and regained by removing the urea. ✓

(B) Function will be disrupted by adding the urea, but it will not be regained by removing the urea.

(C) Function will be gained by adding the urea and disrupted by removing the urea.

(D) Function will be unaffected by the addition and removal of the urea.

Answer A

Correct. Based on the information, adding the urea denatures the enzyme, which will disrupt the enzyme's function by disrupting the active site. In contrast, removing the urea causes the enzyme to refold, which will restore the active site and the enzyme's function.

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139.

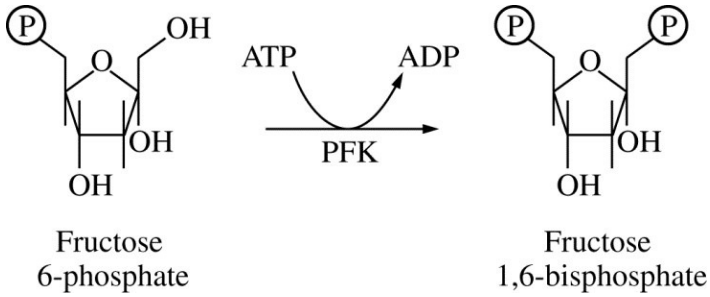


Figure 1. Reaction catalyzed by phosphofructokinase (PFK) during glycolysis

Phosphofructokinase (PFK) is an enzyme that catalyzes the conversion of fructose 6-phosphate to fructose 1,6-bisphosphate during glycolysis, as represented in Figure 1.

PFK can be allosterically inhibited by ATP at high concentrations. Which of the following is the benefit of regulating glycolysis by the concentration of ATP?

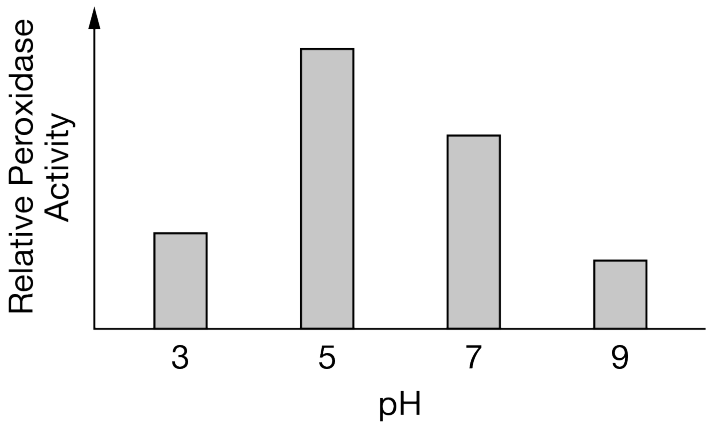
(A) Glycolysis proceeds when the intracellular concentration of ATP is low, which provides ATP to drive cellular reactions. ✓

(B) Glycolysis proceeds when the intracellular concentration of ATP is high and the cell stores ATP for future use.

(C) Glycolysis is inhibited when the intracellular concentration of ATP is low because PFK requires ATP as a substrate for the reaction it catalyzes.

(D) Glycolysis is inhibited when the intracellular concentration of ATP is high because ATP will compete with fructose 1,6-bisphosphate for binding to the active site on the enzyme.

140. The enzyme peroxidase is found in many organisms. It catalyzes the breakdown of hydrogen peroxide into water and oxygen gas. The rate of peroxidase activity at different pH values was assessed by students in the lab. The students' results are shown in graph 1.



Graph 1. Effect of pH on peroxidase activity

If the experiment is repeated at pH 11, the observed activity level of the enzyme will most likely be

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(A) the same as the level at pH 7

(B) lower than the level at pH 9 ✓

(C) greater than the level at pH 9

(D) between the levels observed at pH 5 and pH 7.

Answer B

Correct. Peroxidase activity drops off as the pH increases beyond 5, indicating that the optimal level was closer to pH 5. The enzyme is likely to be further denatured by a pH greater than 9. Since this enzyme occurs in many organisms, it is reasonable that the optimal pH is closer to neutral than to a strongly basic or strongly acidic pH.

141. In plants the enzyme EPSP synthase catalyzes a reaction between two different substrates, S3P and PEP, to form a product used in the synthesis of three amino acids. The structure of glyphosate, the active chemical in many herbicides, is very similar to that of PEP.

Which of the following explains how glyphosate most likely functions as an herbicide?

(A) Glyphosate prevents synthesis of the three amino acids by competing with PEP for binding to the active site of EPSP synthase. ✓

(B) Glyphosate prevents synthesis of the three amino acids by binding to an allosteric site of EPSP synthase, increasing the likelihood that PEP will bind to the active site.

(C) Glyphosate causes too much synthesis of the three amino acids by reducing the activation energy of the reaction catalyzed by EPSP synthase.

(D) Glyphosate alters the structure of the three amino acids by binding to both S3P and PEP in the active site of EPSP synthase.

Answer A

Correct. Because the structure of glyphosate is very similar to that of PEP, glyphosate acts as a competitive inhibitor of the EPSP enzyme by competing with PEP for the active site of the enzyme.